

Tutorial review of Reverse Osmosis and Electrodialysis

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Abstract

Reverse osmosis (RO) and electrodialysis (ED) are the two most important membrane technologies for water desalination and treatment. Their desalination and transport mechanisms are very different, but on closer look also have many similarities. In this tutorial review we describe state-of-the-art theory for both processes focusing on simple examples that are helpful for the non-specialist and for classroom teaching. We describe relevant theory for ion and water transport and the coupling with theory for chemical and mechanical equilibrium on membrane/solution interfaces. For RO of neutral solutes, we explain the solution-friction (SF) model which is closely related to the classical sieving or pore flow model. The SF model includes advection, diffusion, and solute partitioning, and leads to simple relationships for the coupled fluxes of water and solutes (and thus for solute retention as well), also when a diffusion boundary layer is included in the model. Subsequently this theory is extended to describe RO for symmetric salt solutions with charged membranes. For the desalination of salt solutions, both for RO and ED we present two-dimensional module-scale calculations which lead to a characteristic curve that determines optimum operational conditions based on a simple cost calculation that offsets energy and material costs. We discuss the two-fluid model that comprehensively describes ion and water flow both in RO and ED, and we explain how this theory also accurately describes osmosis experiments where water and ions run in opposite directions through a membrane. Finally, we present results of optimization studies of the combination of multiple modules for ED, and we evaluate the relevance of concentration polarization by using a 3D model for cross-current flow in an ED module.

1 Introduction

Reverse osmosis (RO) and electrodialysis (ED) are the two most applied membrane methodologies for water treatment and desalination (deionization) [1,2]. A brief schematic of both methods is presented in Fig. 1. Water treatment generally refers to cleaning water from contaminants other than salts, such as the removal of organic micropollutants (OMPs), whereas desalination and deionization refer to the removal of salts, thus of ions. RO is a method that uses pressure to drive water through a membrane, retaining on the retentate side most of the solutes (solute a generic word referring to all possible molecules and ions in the water), producing freshwater on the permeate side. Nanofiltration (NF) is a companion technology of RO that uses lower pressures, and membranes with larger pore sizes than in RO [3]. In NF the retention of monovalent ions is much lower than of divalent ions and thus divalent ions are selectively removed. In ED, water flows through thin channels next to ion-exchange membranes (IEMs). An applied current pulls the ions from one

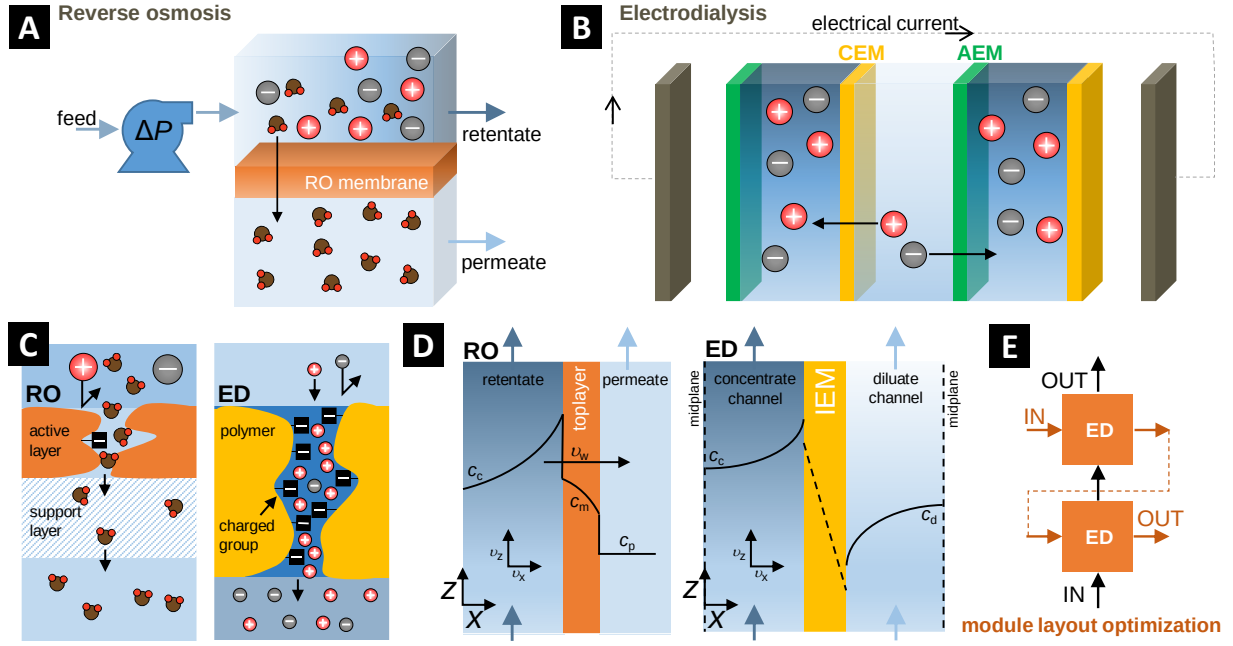


Figure 1: Overview of reverse osmosis (RO) and electrodialysis (ED). A) In RO, a pressure is used to drive water through a membrane, retaining on the retentate side most of the solutes. B) In ED, electrical current runs across channels and membranes, generated in electrodes placed at both ends of a stack that holds many cell pairs. C) Membrane charge and chemical affinity lead to partitioning of solutes on the two outsides of a membrane. D) On the upstream side of an RO membrane (retentate side), ion concentrations increase towards the membrane (concentration polarization) through a diffusion boundary layer. On the downstream side, freshwater (permeate) is produced. Concentration profiles develop in an ED cell pair with diluate and concentrate channels separated by ion-exchange membranes. E) An ED system can consist of multiple modules of different size, which can be connected in various ways.

set of channels through the IEMs. The theories that we present for RO and ED, and the relevant metrics that we define, also apply to NF and EDI and other related desalination methods.

Though ED and RO are very different methods and use distinctly different physical mechanisms, our aim is to demonstrate that the underlying transport theory for flow of water and solutes is the same. As we will show, the effects of charge and current that are of key importance in ED, also play a role in RO. Similarly, in ED pressures develop which influence ion and water transport. Thus it is relevant to present a generalized treatment that applies to both process types. We also demonstrate that key performance indicators on the module level, of relevance for (economic) process optimization, are defined for RO and ED in very similar terms, and relate to one another in a similar fashion.

In this review we focus on the theory of desalination of ionic solutions, i.e., how to use membranes to obtain freshwater from brackish water or seawater by RO and ED. In Section 2 we describe thermodynamic equations and metrics that apply to both methods. We continue in Section 3 with a discussion of simple models for RO that assume neutral solutes, and we show how the solution-diffusion (SD) model can be replaced by the more accurate solution-friction (SF) model. We also present results of a 2D flow calculation for an RO module and derive an analytical equation for the relevant advection-dominated limit. For ED we discuss in Section 4 a co-current plug

		Δc ?	current	ΔP^h ?
Technology	RO/NF	✓	= 0	✓
	ED(I)	✓	✓	~ 0
Phenomenon/ experiment	Osmosis	✓	= 0	~ 0

Figure 2: In reverse osmosis (RO) or nanofiltration (NF), electrodialysis (ED) or electro-deionization (EDI), and in the osmosis experiment, salt concentrations on the two sides of the membrane are different, there is an electrical current, and there is a pressure difference. Even when the current or pressure difference is zero or near zero, that information is still an input in the transport models. Thus, on a theoretical level, these three processes are very similar, and a general theory can be used that describes all three processes.

flow model valid when several system properties are symmetric. Both for RO and ED we set up in these sections a method of cost calculation that demonstrates why the counter-intuitive result that ‘inexpensive membranes have higher energy efficiency’ holds true. In Section 5 we describe the general two-fluid model (TFM) and we apply the resulting equations to a novel data set for the osmosis experiment where we have counter-directional salt and water flow, and we explain the origin of the phenomenon of osmosis: *why* does water flow to a solution of high salinity? In Section 6, the theory for RO and ED that we set up earlier, is extended to describe several highly relevant cases. For RO we discuss in this section theory describing retention of ions and the associated effects of charge and current, and for ED we discuss the effect of flow geometry within the ED stack, and the effect of how multiple units can be stacked together. For ED, we also present results of a 3D cross-current calculation including concentration polarization.

We illustrate in Fig. 2 that RO and ED have many aspects in common with one another, which they also have in common with the phenomenon of osmosis. In all these processes, the same three features play a key role, as we discuss next. First, in all cases there are differences in concentration between the two sides of the membrane. Furthermore, there is a certain electrical current, and finally there is a pressure difference across the membrane. In certain cases, the electrical current will be zero, as in RO, and likewise the hydrostatic pressure difference will be (close to) zero in some cases, for instance in the osmosis experiment when the two sides are open to atmosphere and the liquid levels about equal. Nevertheless, in the transport theory, current and pressure being zero is a key part of the model structure, i.e., when building a theoretical model, it does not matter very much whether current is zero or non-zero, and in all cases a relation for current is part of the model structure. And the same holds for the pressure difference, unless we do not care about pressure, and assume we know the value of the water velocity (zero or otherwise). Because of these similarities in the required theoretical description for RO, for ED and for osmosis, a combined discussion of these processes is important and insightful.

In this review we necessarily can only address a limited number of topics, and must refer the reader to other literature for many other topics of relevance in RO and ED, such as: ion selectivity in multi-ionic mixtures [4], removal of organic micropollutants, module and stack design, fouling abatement, studies of membrane lifetime and robustness, membrane cleaning, the design of spacers and membrane geometries, and full system-level cost optimization.

2 Thermodynamics and Metrics

For all water desalination methods, in all cases the same thermodynamic equations apply, and the same setpoints and metrics (performance indicators) can be used. It is useful to start the discussion with the topic of setpoints and metrics.

2.1 Setpoints and Metrics for water desalination

Every water desalination process is defined by certain operational characteristics, i.e., objectives, or setpoints. These are the amount of water to be treated, the water recovery, and the extent of desalination (or salt removal). Different setpoints can be defined but in the end they describe how much freshwater is obtained from how much feedwater, and what is the salt concentration of the product water. These are considered as setpoints and are ‘hard’ objectives that must be attained.

After the setpoints of a process have been defined, metrics (or, equivalently, ‘performance indicators’) come into play. They describe optimized operation. Such metrics can be manifold. We can for instance have a measure of energy use (energy efficiency), the degree of removal of specific compounds (selectivity), or a measure of how well the membrane is retaining its integrity over time and able to handle fouling.

2.1.1 Setpoints for water desalination

Setpoints for a water desalination process are operational parameters that are ‘hard’, that must be attained. It is not the case that they are just ‘nice to have’, i.e., to be optimized in some direction. Instead, they *define* a certain desalination operation. They are also essential for a calculation of the theoretical minimum energy use (see Section 2.2). In practice, the aim is not that these setpoints are reached in an individual desalination unit (module, stack), but they will be the result of operation of a total desalination plant. So a first battery of units may desalinate to a certain percentage, and only in a next step the final salt concentration is reached. In that case, setpoints are only reached at the overall level of a full water desalination plant.

The first setpoint that defines operation of a desalination plant or module, is how much freshwater (diluate, permeate) is obtained in m^3/s , i.e., the freshwater flowrate, $\phi_{v,p}$ (where ‘p’ refers to permeate or product water). Now, even at this most basic level, caution is needed, because operation is generally not ‘24/7’, i.e., not all-day, all-year-round. Instead, any plant will –for one reason or another– change operational conditions during the day or year, for instance running only when electric energy is cheap. So even for the setpoint of productivity, it must be defined if this is ‘peak productivity’, or a value averaged over many days or longer. In this review we neglect such operational considerations, and focus on steady-state operation of one or two modules, with constant flow rates and setpoints.

The second setpoint is water recovery, WR, which is the fraction of all feedwater that is turned into freshwater, thus

$$\text{WR} = \frac{\phi_{v,p}}{\phi_{v,p} + \phi_{v,c}} = \frac{\phi_{v,p}}{\phi_{v,f}} \quad (1)$$

where $\phi_{v,j}$ ’s are volumetric flowrates in m^3/s . Other terminology is recovery ratio, often described by the symbol r , RR , or α . All flowrates $\phi_{v,j}$ can be replaced here by ‘integrated’ volumes, V_j . Subscripts ‘f’, ‘p’ and ‘c’ refer to the feedwater, product water (freshwater, or in RO: permeate), and concentrate (retentate), respectively. A related definition is ‘split ratio’ for the ratio $\phi_{v,p}/\phi_{v,c}$, which equals $\text{WR}/(1-\text{WR})$.

The third setpoint is the extent of desalination, which one can define as the difference in salt concentration between feedwater and product water, $\Delta c = c_f - c_p$. Alternatively, the setpoint can simply be the salt concentration of the freshwater that is obtained. Another way to express this setpoint is as a retention, rejection, or passage. Salt retention, R_i , is defined as

$$R_i = 1 - c_{p,i}/c_{f,i}. \quad (2)$$

Retention and rejection have the same meaning. The passage of a solute is P_i and is given by $P_i = 1 - R_i$. These latter definitions of R_i and P_i are particularly useful for RO, not so much for ED. The above three setpoints can be rewritten to other setpoints, but the above formulations are common choices.

In general, we can attain the second and third setpoints in any well-designed RO or ED module, but must then adjust the flowrates of the water treated. Besides the flowrates, in ED we can tune the current density (or voltage) that is applied to a stack. So in most cases, it will be possible to find an operational setting where the second and third setpoints are attained, for a given flow of water that is treated. The same is valid in RO, where we have two flow rates and a pressure difference that can be tuned.

The third setpoint, the degree of desalination, relates to a total or average desalination of the feedwater as a whole. In reality, most water sources contain many different salts, and we aim for a certain separation level of several critical salts amongst all salts. In practice, still, it is useful to treat the desalination as a setpoint, for instance as a ‘total Δc ’ formulated as a summation over all ions, including perhaps a weighing to account that some ions are more relevant to remove than others. The degree of desalination for individual ions, related to ion selectivity, this factor we relegate to the next category, of a metric.

Thus, for a given module, we can arrive at setpoints for WR and Δc , but this can only be achieved for one (maximum in the) flowrate of the water that is treated. I.e., we cannot just achieve any level of the volumetric rate of freshwater that one may desire in a given module, if there are two or more other setpoints to achieve. This can also be rephrased into the statement that the setpoints for WR and Δc determine the ratio of system size (e.g., membrane area) over volume flowrate of the water to be treated (or, to be obtained as freshwater). Let us call this factor the specific membrane area $\mathcal{A}$, i.e., the total membrane area per m^3/s of freshwater to be obtained. Typically, with more stringent setpoints for WR and Δc (i.e., higher WR and Δc), this factor $\mathcal{A}$ will increase, i.e., for a certain volume of freshwater to be obtained, the system size will have to increase (more membrane area). In addition, also more energy will be required to attain higher WR and Δc . We will quantitatively analyze these predictions in upcoming sections on RO and ED. For now, we make a more simple analysis. When we know the factor $\mathcal{A}$, i.e., the system-size-per-volume-flow-of-freshwater, then a simple calculation gives us the total system size (membrane area, or number of modules) that must be installed for a certain volumetric freshwater flowrate. Thus, for instance,

$$\text{total membrane area in an installation} = \mathcal{A} \cdot \phi_{v,p}. \quad (3)$$

Hence, the size of a certain system to be installed, specifically the membrane area, depends linearly on the required (peak) capacity, $\phi_{v,p}$, and depends linearly on the factor $\mathcal{A}$, which is the inverse of a system ‘compactness’: a compact system is one that can do a lot for a certain membrane area.

Often the primary aim is not so much the membrane area, or system size, but the financial costs of installing and running a desalination plant. We can imagine that these costs (both as investment, and yearly replacement) depend linearly on the price of membranes, because membranes are one of the critical components of a desalination plant. We then arrive at

$$\text{costs of operation} = \text{a.m.u./m}^2 \cdot \mathcal{A} \cdot \phi_{v,p}. \quad (4)$$

where a.m.u./m² is the membrane purchase price in an arbitrary monetary unit (a.m.u.) per m². Thus, how to keep down the investment and operational costs of a desalination plant? Three factors can already be identified that must then be minimized: the purchase price of a square meter of membrane, the specific system size $\mathcal{A}$ (i.e., system efficiency or compactness, which is $\mathcal{A}^{-1}$), and how much water needs to be produced. In reality there are many more factors of concern, including lifetime of membranes, costs of maintenance, cleaning and replacement, and energy costs [5]. Beyond membranes, there are multiple other cost factors related to plant operation, a discussion of which goes beyond the scope of this review.

2.1.2 Metrics for water desalination

Metrics, or performance indicators, are those factors that for a certain desalination process, defined by certain setpoints, are ‘nice to have’, that have a value that ideally is as high, or as low, as possible.

A first metric tracks the specific removal of certain key compounds, and the more the removal of these, the better. So a metric could be a removal degree of one or several key ions, perhaps expressed as single number by summing over several key ions including a weighing factor describing the importance of the removal of each.

Another example of a metric is the energy efficiency, η , which is the ratio of the theoretical minimum energy to achieve a certain desalination (defined by the setpoints) over the actual energy investment. [This metric is also called TEE, for ‘thermodynamic energy efficiency’.] There are many possible definitions for η , depending on which energy contributions are included in the actual energy input. For instance, for ED we can limit analysis to the electrical energy input required for current to flow across the stack, but we can also include the (much smaller) costs of pumping the water through the channels. In RO we can decide to include for a certain percentage the recovery of ‘pressure energy.’ We can take into account efficiencies of power sources and pumps, and of other electric equipment, for instance to convert (AC) electricity into pumping power, or into DC stack current [6]. So, a report detailing values of η must carefully lay out which energy factors are considered.

Often, instead of an efficiency η , the inverse factor is reported, which then is the factor by which the actual desalination energy is ‘above the minimum.’ Generally, values as low as two, for this inverse factor, are considered to be very low, i.e., we now have an actual energy use that is only a factor two above the thermodynamic minimum. In many cases, especially for water of low salinity, this factor is much higher, for instance 5 or 10, which implies the actual energy is equal to that number of times the thermodynamic minimum.

Other metrics may refer to lifetime of the membranes and track the state of membrane functionality and integrity over time. An overall metric is operational costs. So for given setpoints (i.e., objectives of the desalination process), this metric tracks the success of operational management to minimize the sum-total of yearly operational cost factors, including energy, membrane replacement, and other contributions. It is clear that many metrics can be defined and used. But important is that the objectives are clearly set, and metrics clearly defined, when comparing variations within a certain technology, let alone when making comparisons between different desalination methods.

2.2 Minimum energy of desalination

For given setpoints, we can calculate the theoretical minimum energy that must be invested to attain that particular degree of desalination. This minimum energy does not depend on what method is used to desalinate. Of course, both in RO and ED there are additional constraints that increase the minimum energy further: for instance, in an RO module we can only apply one pressure, which –on that side of the membrane– is then effective through the entire module. And in an ED stack, there is one value of the cell pair voltage, which then applies from inlet to exit of each channel. All kinds of modifications are possible to relax these system limitations, especially by working with multiple modules or stacks that operate at different conditions. But still, the thermodynamic minimum that we will analyze in this section, that is system-independent, and only depends on the composition of the water to be treated, temperature, and the process setpoints.

This minimum energy that must always be invested to desalinate water, $e_{\min}$, can be calculated on the basis of a thermodynamic analysis of the energy in the three streams that are involved. These three streams are the feedwater ‘f’, the obtained freshwater (permeate, ‘p’, or diluate, ‘d’), and the concentrate (or, retentate, ‘c’). In each stream there are multiple ions, each at a concentration c_i , and all ions are used in the general equations we will discuss first. For a 1:1 salt of one cation and one anion, we can define a salt concentration, c_{∞} , which is equal to c_i of each of the two ions involved.

2.2.1 The thermodynamics of ideal solutions

For thermodynamically ideal solutions, the only contribution to the theoretical minimum energy of desalination, $e_{\min}$, is the decrease in ion entropy (Gibbs free energy of mixing), and thus $e_{\min} = -T\Delta S$ where

$$\Delta S = S_p + S_c - S_f \quad (5)$$

with the entropy of each stream given by

$$S_j = -R \sum_i \phi_{v,j} c_{i,j} \ln c_{i,j} \quad (6)$$

where R is the gas constant (8.3144 J/mol/K), $\phi_{v,j}$ the volumetric flowrate of a certain stream j (in m^3/s), and $c_{i,j}$ the concentration of ion i in stream j , with $c_{i,j}$ having the unit mol/m^3 , i.e., mM.

This equation can be used for any mixture of ions, and irrespective of how successfully each ion is separated in a membrane process. The complete calculation also includes two balances, first a volumetric balance, which is

$$\phi_{v,f} = \phi_{v,p} + \phi_{v,c} \quad (7)$$

and secondly a mass balance per solute type,

$$\phi_{v,f} c_{f,i} = \phi_{v,p} c_{p,i} + \phi_{v,c} c_{c,i} \quad (8)$$

A calculation based on these equations is easily performed in spreadsheet software, and provides the energy in $\text{W}=\text{J}/\text{s}$ to attain a certain separation in a steady-state process. It can be helpful to interpret $\phi_{v,j}$ as volumes V_j , such that the equation provides the energy in J to treat a certain volume of water. We can define the minimum energy per m^3 of freshwater produced, $E_{\min}$, as $E_{\min} = e_{\min}/\phi_{v,p}$. An important result (to check one’s calculation) is that for a 1:1 salt, with 50% water recovery, i.e., $\text{WR}=0.50$, a feed salt concentration $c_{\infty}=525$ mM, $c_{\infty,p}=0$ mM, and temperature $T=298$ K, that $E_{\min}=1.0$ kWh per m^3 freshwater produced. Note that a 1:1 salt solution with salt concentration c_{∞} contains both cations and anions at that concentration, and both ions must be counted. The ion’s charge does not play a role in calculating the entropy of desalination.

2.2.2 Extensions for non-ideal solutions

We will discuss various simplifications in a later section, but let us first briefly point out how more energy terms can be included in addition to the ion entropy described by Eqs. (5) and (6). This includes first of all the effect of electrostatic Coulomb interactions between ions, which for a 1:1 salt can be described by the modified Bjerrum equation for the contribution of this effect to the free energy density of a solution [7]

$$f_{\text{elec}} = -\frac{3}{2}k_{\text{B}}T\lambda_{\text{B}}n_{\infty}^{4/3}. \quad (9)$$

This term can be evaluated for each of the three streams, and then multiplied by the volume flow rate, ϕ_v , and added to e_{min} (with an additional minus-sign for the feed stream). The salt concentration in numbers per volume, n_{∞} , relates to salt concentration c_{∞} in moles per volume according to $n_{\infty} = N_{\text{av}}c_{\infty}$, where N_{av} is Avogadro's number. Furthermore, λ_{B} is the Bjerrum length which in water at room temperature is around $\lambda_{\text{B}} \sim 0.72$ nm, and k_{B} is Boltzmann's constant. We can rewrite Eq. (9) to

$$f_{\text{elec}} = -\alpha c_{\infty}^{4/3} \quad (10)$$

where at room temperature the prefactor is $\alpha \sim 225 \text{ J} \cdot \text{m}/\text{mol}^{4/3}$. For salt mixtures other than 1:1 salts, numerical analysis of the modified Bjerrum theory is required [7].

Another contribution to the energy is an ion volume (size) effect that shows up at higher concentrations, which describes that the (hydrated) ions cannot overlap, i.e., they exclude space for one another. The Carnahan-Starling equation of state can be used to describe this effect (when we assume that all ions have the same hydrated size and all are more or less spherical), and this leads to a further contribution to the free energy, thus to e_{min} . For each flow this contribution is

$$\frac{f_{\text{vol}}}{RTc_{\text{tot}}} = \frac{4\phi - 3\phi^2}{(1-\phi)^2} = 4\phi + 5\phi^2 + 6\phi^3 + \dots \quad (11)$$

where c_{tot} is the total ion concentration (for a $z:z$ salt, $c_{\text{tot}} = 2c_{\infty}$) and ϕ is the volume fraction of all ions together, i.e., $\phi = c_{\text{tot}}v_i$, where v_i is the molar volume of an ion. The contribution f_{vol} can be multiplied by ϕ_v and added to e_{min} . As a numerical example, for a 1:1 salt at $c_{\infty} = 525$ mM, with ions of a hydrated size of $\sigma = 0.5$ nm (size is twice the radius), the ion volume fraction is $\phi = 0.0414$. With the electrostatic and volumetric contributions implemented, then for the previous numerical example, the minimum energy consumption increases from $E_{\text{min}} = 1.00 \text{ kWh}/\text{m}^3$ to $E_{\text{min}} = 1.14 \text{ kWh}/\text{m}^3$. Further extensions relate to the energies of ion-ion association and ion (de-)protonation [8], but we do not discuss these effects in this review.

2.2.3 Simplifications of the thermodynamic expressions

Various simplifications are possible to the above general expressions for the theoretical minimum energy. Here we consider a symmetric $z:z$ salt solution, with one type of cation and one type of anion. We only discuss the ideal, entropic, contribution. We present results for the energy per m^3 of permeate (diluate, freshwater) produced. Each of the three streams, feed, permeate and concentrate, is described by a certain value for the salt concentration c_j . The salt mass balance

$$c_{\text{f}}\phi_{v,\text{f}} = c_{\text{p}}\phi_{v,\text{p}} + c_{\text{c}}\phi_{v,\text{c}} \quad (12)$$

is included in the analysis, together with the volumetric balance, Eq. (7). A first simplification relates to RO in the limit of very low WR and very low c_{p} . In this limit, on the feed side the concentration does not change along the module. We only produce a very low volume of totally

clean freshwater. We now obtain the result that the minimum energy per volume of freshwater is given by

$$E_{\min} = \Pi_f \quad (13)$$

where Π_f is the osmotic pressure of the feed solution. The osmotic pressure for ideal conditions is given by $\Pi = RT \sum_i c_i$, and thus for a $z:z$ salt, $\Pi = 2RT c_\infty$.

In general, for a perfectly selective membrane, and thus $\Pi_p = 0$, the hydrostatic pressure difference that must be applied across the membrane in a pressure-driven process, $\Delta P^{h,\infty}$, must be larger than $\Pi_{\text{dbl/mem}}$, where $\Pi_{\text{dbl/mem}}$ is the osmotic pressure at the interface between the diffusion boundary layer (DBL) and membrane, and this osmotic pressure is larger than Π_f by the over-pressurization factor, α_1 . This effect is commonly called concentration polarization (CP). The factor α_1 can be close to unity, when CP is negligible, which will be the case for a low permeate flux and high enough crossflow of water (water flow along the membrane), but values of $\alpha_1 = 1.5$ or larger are also possible. The hydrostatic pressure is typically larger than $\Pi_{\text{dbl/mem}}$ by a second factor, α_2 , again larger than 1, to have transmembrane water flow. Thus, in this simplest analysis of a perfectly selective membrane at $WR \rightarrow 0$, the energy efficiency of an RO membrane is given by

$$\eta_{\text{RO}} = \frac{\Pi_f}{\Delta P^{h,\infty}} = \frac{1}{\alpha_1 \alpha_2}. \quad (14)$$

In a more general case, combining the various balances with the expressions for S_i , we obtain for $E_{\min}$, now for arbitrary values of WR and c_p , for a $z:z$ salt of one anion and one cation, the result that [9]

$$\frac{E_{\min}}{2RT} = \frac{c_f}{WR} \ln \frac{c_c}{c_f} - c_p \ln \frac{c_c}{c_p} \quad (15)$$

where all concentrations c_i are salt concentrations in mol/m^3 . When c_p approaches zero, i.e., complete salt removal, then Eq. (15) simplifies to [9]

$$\frac{E_{\min}}{2c_f RT} = -\frac{\ln(1-WR)}{WR} = 1 + \frac{1}{2}WR + \frac{1}{3}WR^2 + \dots \quad (16)$$

which in the limit of $WR \rightarrow 0$ simplifies to Eq. (13). Note that if we apply these equations to a solution containing a single neutral solute, not a 1:1 salt, then factors ‘2’ on the left side of Eqs. (15) and (16) must be omitted.

It is very important to note that Eqs. (13) and (16) are only correct when $c_p \rightarrow 0$. For non-zero values of c_p , this energy, in the limit of $WR \rightarrow 0$, is given by

$$E_{\min}/2RT = c_f - c_p - c_p \ln \frac{c_f}{c_p}. \quad (17)$$

2.3 Practical energy use in RO and ED

In the above section the thermodynamic, minimum, energy of a desalination process was discussed. But what is the actual, real, or practical, energy consumption? The main factors are the pressure required to pump fluid through channels, and in RO to press water through the membrane, while in ED we need energy to generate a flow of electrical current across the ED stack.

The energy required to pressurize a stream is ΔP^h times flowrate $\phi_{v,j}$, which is a number with unit $\text{J/s} = \text{W}$. This is the energy to pump feedwater into a module. And to pump water exiting one module into the next operating at a higher pressure. At the end of a train of modules, it is possible to install an energy recovery device (ERD). If the water on the retentate side (the concentrate; a smaller volume than the flow of feedwater) passes an ERD, exiting at atmospheric pressure, an

energy can be recovered equal to $\eta_{\text{ERD}} \Delta P^h \phi_{v,c}$, with the energy efficiency of the ERD possibly as high as $\eta_{\text{ERD}} = 0.8$ when sufficient volumes are treated. If the volume of retentate is small (i.e., the water recovery is high), then installation of a pressure recovery device is generally not economically attractive.

Another practical limitation, even for ideal operation, is that we generally have a limited number of modules. In RO with a single module, for a membrane that perfectly retains all solutes, the minimum pressure is given by the retentate (concentrate) concentration, which in this case for perfect retention relates to c_f by $c_c = c_f/(1 - \text{WR})$. So the minimum energy per unit permeate, practically, cannot be less than $e_{\text{min,prac}} = RT c_f/(1 - \text{WR})$ [9], and with E_{min} given by Eq. (16), this implies a practical single stage RO efficiency of

$$\eta_{\text{RO}} = \frac{1 - \text{WR}}{\text{WR}} \cdot \ln(1 - \text{WR}) = 1 - \frac{1}{2} \cdot \text{WR} - \frac{1}{6} \cdot \text{WR}^2 - \dots \quad (18)$$

which can be multiplied by $1/\alpha_1\alpha_2$.

In ED also some energy is required to push water through tubings and channels of the ED stack, but this is low compared to the main energy input, which is electricity for running current across the stack. With current I (in A, i.e., Ampère), and a cell pair voltage V_{cp} , then for each cell pair the electrical energy input is $I \cdot V_{\text{cp}}$ (unit J/s=W), and with N cell pairs in an ED stack, this energy is to be multiplied by N to obtain the electrical energy input for the stack. Inside each cell pair the voltage V_{cp} is due to resistances for current to cross the diluate and concentrate channels, to cross the membranes, and due to the Donnan voltages at the membrane-solution edges. These Donnan voltages are not a resistance (they are not a direct function of the current), but are a direct consequence of the thermodynamics of desalination, i.e., the Donnan voltage cannot be reduced to zero by running at low currents. Of the Ohmic resistances, the largest loss is for the current to cross the low-salinity diluate channel. In EDI this resistance is reduced by filling these channels with ion-exchange resin material. In an ED stack (of 10s or 100s of cell pairs) there are also two end-compartments in which electronic current transfers to ionic current, and vice-versa. There is a voltage loss associated with the Faradaic reactions there, but with increasing numbers N of cell pairs, this voltage loss (of the order of the voltage over 1 cell pair) becomes inconsequential for the energy requirements of a full ED stack. The effect of single-stack limitations in ED, because generally only a single value of the cell voltage is applied to the stack as a whole, are discussed in Section 4.

The energy thus invested in RO and ED by pressure and electricity (including energy recovery), is always larger than the thermodynamic energy as calculated in Section 2.2. Thus, the ratio of thermodynamic energy e_{min} over the real, or practical energy, i.e., the energy efficiency η , is always less than unity, i.e., we always have $\eta < 1$.

3 One- and two-dimensional models for reverse osmosis with neutral molecules

3.1 Flowrates and retention in a 1D model of RO for neutral solutes

Reverse osmosis (RO) is a process to remove solutes from water by pushing water through a membrane which largely blocks passage for solutes. This is the general mechanism of all pressure-driven membrane processes, but for water desalination the size of the pores (free volume) in the membrane must be very small, typically well below 1 nm in the selective toplayer of an RO membrane. In classical literature, RO is called hyperfiltration, but nowadays this term has been re-

placed by RO. Nanofiltration (NF) is similar to RO but has slightly larger pores, and thus the applied hydrostatic pressure is lower, with the retention of monovalent ions significantly reduced. NF is therefore suitable for the selective separation of divalent ions and other larger molecules from monovalent ions, because these species are still rejected at a high level.

Because RO (and NF) can be used not only to desalinate water but also to remove other (larger) molecules that may not necessarily be charged, it is useful to start with theory for uncharged solutes. A key parameter is the transmembrane water velocity, v_w , also called ‘filtration velocity’ [4]. This is the velocity of water in the direction across the membrane. For uncharged solutes, if v_w is known, and if there are no special interactions between the solutes, then, interestingly all solutes can be treated separately, and their transport rates are not coupled. This is also why equations that we derive below for the retention of solutes in RO are independent of feed solute concentration, as long as v_w is set to a certain value. If v_w is not fixed, but instead the applied pressure is set to a certain value, then this is not the case but via the osmotic pressure the concentration ‘feeds back’ into the transport problem and into the (prediction of) retention. Note that this situation is very different for electrolyte solutions (i.e., a solution containing ions), because then an electric field develops to ensure that the electrical current is zero, and this always leads to various couplings between the ion fluxes. RO for electrolytes will be discussed in Section 6.

Based on a solute-water-membrane friction model, see Section 5, we can derive for the flux of neutral species across a porous membrane

$$J_i = K_{f,i} c_{m,i} v_w - K_{f,i} D_{m,i} \frac{\partial c_{m,i}}{\partial x} \quad (19)$$

where the transmembrane solute flux J_i and water velocity v_w are defined per unit geometric membrane area, thus they are superficial velocities, i.e., they are not pore-based, interstitial, velocities. Concentrations are defined per unit pore volume, i.e., per volume of the open water-filled pores in the membrane. Eq. (19) includes transport due to a solute concentration gradient (second term), as well as advection of solutes, which is the movement of solutes because they ‘flow along’, i.e., are advecting with, the water that flows through the membrane (first term). Here, x is the coordinate axis directed across the membrane. The diffusional term depends on a membrane-based diffusion coefficient, $D_{m,i}$, which includes the effects of porosity and tortuosity. In addition, solute-membrane friction leads to a non-unity hindrance or friction factor $K_{f,i}$, which is in front of both terms in Eq. (19). This equation follows as a simplification of the two-fluid model (TFM) that will be explained in Section 5. The friction factor $K_{f,i}$ depends on the extent of solute-membrane friction, and thus relates to membrane pore size and the (hydrated) size of the solutes. In the absence of solute-membrane friction, the solutes move freely with the fluid (i.e., with the water in the pores, though of course they are still subject to diffusional forces, and electrical forces when charged), and in that case $K_{f,i} = 1$.

Now, for RO, a value of $K_{f,i} = 1$ is unlikely, because that would mean no ion-membrane friction, and thus we expect $K_{f,i}$ to be less than unity. However, it is not likely that we have $K_{f,i} = 0$, in which case not only the advection term cancels (which would lead to the solution-diffusion model that we discuss later on), but then there is no diffusional transport of solutes either, and such a membrane would therefore perfectly block solutes under all conditions. This is not the case in a real RO or NF membrane, with solutes still passing the membrane, and thus it must be the case that $K_{f,i}$ is larger than zero. Thus, in an RO membrane with neutral solutes, diffusion and advection are to be considered jointly, as described by Eq. (19), and it is incorrect to neglect advection altogether.

Before continuing with a discussion of transport models, we must first introduce two other model elements. First, the solute concentration just in the membrane is related to that just out-

side. To that end we make use of a partitioning equation [10,11], which follows from chemical equilibrium of a species i across an interface, which applies on both outer surfaces of a membrane [4], resulting for neutral solutes in

$$\Phi_i = S_i = \frac{c_{m,i}}{c_{\infty,i}} \quad (20)$$

where index ∞ refers to a position just outside the membrane. For RO with ions, see sections 5.2 and 6.1 an additional Donnan effect arises, which in this paper we treat separately, i.e., do not include in S_i or Φ_i . The partitioning coefficient, Φ_i , is the same as the solubility of a species in a material, S_i , and this term is for instance due the chemical affinity of a solute with a certain phase (medium) relative to its affinity with another phase, but also other effects of solute-membrane interaction can be absorbed in Φ_i [10]. In RO with neutral solutes, we always have $\Phi_i < 1$ or $K_{f,i} < 1$.

A further element in RO modeling is the ‘permeate equation’, an equation essential to describe RO in a one-dimensional (1D) geometry, valid when there is no flow along the membrane on the permeate side, which is (Eq. (24) in ref. [12], Eq. (8) in ref. [4])

$$c_{p,i} = \frac{J_i}{v_w} \quad (21)$$

which looks very simple, but is actually a quite intriguing result. It states that the concentration of solutes on the permeate side is a direct function of the ratio of solute flux through the membrane over the water velocity. This is a correct result in steady-state, thus with conditions on the upstream side unvarying in time. It is also valid when on the upstream side conditions change in time and the permeate is regularly removed; as a consequence water that permeated the membrane at earlier moments does not mix up with fresh permeate. In a complete (2D) module, Eq. (21) can also be used with average values of J_i and v_w (averaged over the complete module).

Let us discuss the solution-diffusion (SD) model. In the SD-model, the advection term in Eq. (19) is set to zero, solute-membrane friction is neglected, thus $K_{f,i} = 1$, and thus only diffusion drives solutes across the membrane [13]. (As a side-note, in the SD-model water transport is described in the same way, as if water molecules molecularly diffuse across the membrane.) We can then integrate Eq. (19) on the basis of assuming that the molar flux of solutes, J_i , is constant across the membrane, and after combination with Eq. (20), we arrive at the SD-model

$$J_i = k_{m,i} \Phi_i (c_{f,i} - c_{p,i}) \quad (22)$$

where the membrane mass transfer coefficient, $k_{m,i}$, is given by $k_{m,i} = D_{m,i}/L_m$, with L_m membrane thickness. The abbreviation SD refers to the product of solubility (the Φ_i -term) and the diffusion of solutes across the membrane, described by $k_{m,i}$. In this SD-model, the flow of solutes, and that of the solvent, ‘are completely independent without any effect of one on the other when, in general, they may be coupled by either frictional or convective effects’ [14].

Combining Eqs. (21) and (22) leads to an expression for retention R_i in the SD-model given by

$$R_i = \frac{Pe_i}{Pe_i + \Phi_i} \quad (23)$$

where the membrane Péclet-number is given by $Pe_i = v_w/k_{m,i}$. Eq. (23) predicts that if we steadily increase the transmembrane water velocity v_w , thus Pe_i , at some point we always reach a retention as close to unity as desired [15]. However, this seems unrealistic, that this ideal limit can always be reached, irrespective of membrane properties. That the SD-model leads to the prediction of a 100% retention at high water velocities v_w , irrespective of membrane properties, is generally not apparent because Eq. (22) is often combined with a model for the diffusion boundary layer (DBL)

(often by a transport equation that does include both advection and diffusion), and then a model is arrived at where retention will be less than unity, but with the origin of this non-unity maximum in retention now due to a mass transfer limitation in the DBL. We agree with the general observation that retention must be able to stay below unity also at high v_w , and we agree that concentration polarization is important. However, we must also make sure that the correct origin of a non-unity retention is identified, and that the membrane is modelled correctly. We argue that such a correct membrane model must include an advection term, and that improved model we analyze next.

We continue with an RO transport model for neutral solutes, which simultaneously includes the effects of solute partitioning, advection, solute-membrane friction, and the DBL. Calculation results stemming from this model are presented in Fig. 3. If we assume steady-state, and one-dimensional flow, solute flux J_i is the same at each position in the membrane, and we can integrate Eq. (19) across the membrane to obtain

$$J_i = K_{f,i} v_w \frac{c_{m,L,i} \exp(\text{Pe}_i) - c_{m,R,i}}{\exp(\text{Pe}_i) - 1} \quad (24)$$

where subscripts L and R refer to the left and right sides of the membrane, at positions just inside the membrane. We associate ‘left’ with the feed/retentate side, for which we use the symbol ‘f’ in this section, and ‘right’ with the permeate side (freshwater, diluate), for which we use subscript ‘p’. Eq. (24) is sometimes referred to as the Hertz equation [16].

We combine Eqs. (20) and (24), which leads to

$$J_i = K_{f,i} \Phi_i v_w \frac{c_{f,i} \exp(\text{Pe}_i) - c_{p,i}}{\exp(\text{Pe}_i) - 1} \quad (25)$$

which is the general result for retention in an RO membrane with neutral solutes (e.g., Eq. (10) in ref. [17]). Because of the dependence of solubility, i.e., partitioning, described by the factor Φ_i , in combination with ion-membrane friction, captured by the factor $K_{f,i}$, we call this the solution-friction model, or SF-model. It is identical to the ‘sieving model’ put forward by Kedem and colleagues in the 1950s and 1960s, though there are differences in the derivation. Starov and Churaev [4] arrive at the same expressions and call this the ‘homogeneous model’ for RO membranes. Interestingly, we can derive that the sieving coefficient σ_i relates to $K_{f,i}$ and to Φ_i according to

$$\sigma_i = 1 - K_{f,i} \Phi_i. \quad (26)$$

This result was already arrived at by Spiegler and Kedem, see their Eq. (49) in case in their equation we implement that $v_s \ll v_w$, see also Eq. (76) in ref. [18]. We combine Eqs. (2), (21), and (25) to obtain for solute retention

$$R_i = 1 - \left[1 + (\sigma_i^{-1} - 1)^{-1} (1 - \exp(-\text{Pe}_i)) \right]^{-1}. \quad (27)$$

For low v_F , the expression for solute flux J_i simplifies to

$$J_i = (1 - \sigma_i) (k_{m,i} (c_{f,i} - c_{p,i}) + \langle c \rangle v_w) \quad (28)$$

where $\langle c \rangle$ is the average of the outside solute concentrations on either side. Eq. (28) is called the arithmetic mean expression [16]. Retention in this limit is given by

$$R_i = \frac{\sigma_i}{1 - \sigma_i} \frac{v_w}{k_{m,i}} \quad (29)$$

which shows that for very low water velocities, retention goes to zero, and for a non-zero retention we need $\sigma_i > 0$ and thus we need either a non-unity $K_{f,i}$ -factor or a non-unity partitioning function, Φ_i . In the other limit, of a high transmembrane water velocity, v_w , Eq. (25) simplifies to

$$J_i = (1 - \sigma_i) c_{f,i} v_w \quad (30)$$

which implies that in this limit solute flux is only due to advection, and no longer a function of diffusion. Thus in this limit solutes flow across the membrane because they are dragged with the water (i.e., the mechanism of advection), without any significant diffusional effect, as already proposed in ref. [19]. We call this the advection-dominated regime or advection-dominated limit. Eq. (30) is strikingly different from the SD-model, Eq. (22), which neglects advection altogether and instead assumes that only diffusion is a driving force for solutes to transfer across the membrane. In this high-Pe limit, retention is now given by

$$R_i = \sigma_i \quad (31)$$

which is Eq. (30) in ref. [12]. Eq. (31) shows how any RO membrane has a natural limit in what retention it can achieve, determined by the extent to which solutes are excluded from the membrane (which implies a value of $\Phi_i < 1$), and by the extent of solute-membrane friction (the more the better, which leads to $K_{f,i} < 1$). If a membrane does not do either, i.e., it does not exclude solutes, i.e., $\Phi_i = 1$, and it does not impose a frictional force on solutes at all, i.e., $K_{f,i} = 1$, then the SF theory predicts that retention will be zero, and in our view that is a correct result: for an RO membrane to function, either solutes must be excluded from the membrane, leading to $\Phi_i < 1$, or there must be a solute-membrane friction, i.e., $K_{f,i} < 1$, and ideally both. Instead, the SD-model, Eq. (23), does not make these detailed predictions at all. A similar discussion is in ref. [14]. Thus, if neither of these two factors plays a role, the SF model predicts that retention is zero at all values of v_w , and this is different from predictions of the SD-model. Thus Eqs. (27) and (31) are correct model equations, to replace Eq. (23).

Interesting is the influence of membrane thickness, which influences $k_{m,i}$. In the limit of a high transmembrane water velocity there is no effect of thickness on retention, but there is for lower values of v_w . In that case a thicker membrane leads for the same v_w to more retention. Or vice-versa, when the membrane is made thinner, at the same v_w retention decreases, because diffusion of solutes increases. Indeed for ever lower thicknesses, thus lower Pe_i , the limiting behaviour of Eq. (27), given by Eq. (28), is that retention goes to zero, see Fig. 3A. Thus, thin(ner) membranes are not necessarily good.

3.2 A 1D model for RO of neutral solutes including concentration polarization

Next we extend the model for a membrane with a second model element which represents solute transport through a diffusion boundary layer (DBL), or alternatively called concentration polarization layer (CP layer) [20]. This layer is assumed to have a certain thickness and all water and solutes are to be transported through this layer before reaching the membrane. Thus it is located in front of the membrane, i.e., on the upstream side. Eq. (24) which describes solute advection and diffusion in a membrane also applies to the DBL, now with $K_{f,i} = 1$ because there is no porous structure in the DBL with which solutes can have friction, while $\Phi_i = 1$ applies to the solution-DBL ‘interface’. When we solve the DBL structure jointly with membrane transport, then we must also calculate the concentration in the DBL at the DBL/membrane interface and relate that to the concentration just in the membrane by the partitioning equation, Eq. (20). For neutral solutes, this concentration at the DBL/membrane interface, just outside the membrane, is obtained from rewriting Eq. (24) (implementing $K_{f,i} = 1$, $Pe_i \rightarrow Pe_{dbl,i}$, and $c_{m,L,i} \rightarrow c_{f,i}$), resulting in

$$c|_{\text{just outside membrane}} = \frac{J_i}{v_w} + \left(c_{f,i} - \frac{J_i}{v_w} \right) \cdot \exp(Pe_{dbl,i}) \quad (32)$$

where $Pe_{dbl,i} = v_w/k_{dbl,i}$, in which $k_{dbl,i} = D_{\infty,i}/L_{dbl}$ where L_{dbl} is an estimated thickness of the DBL layer. With more stirring, or a higher cross flow rate (flow rate of the water pumped along the

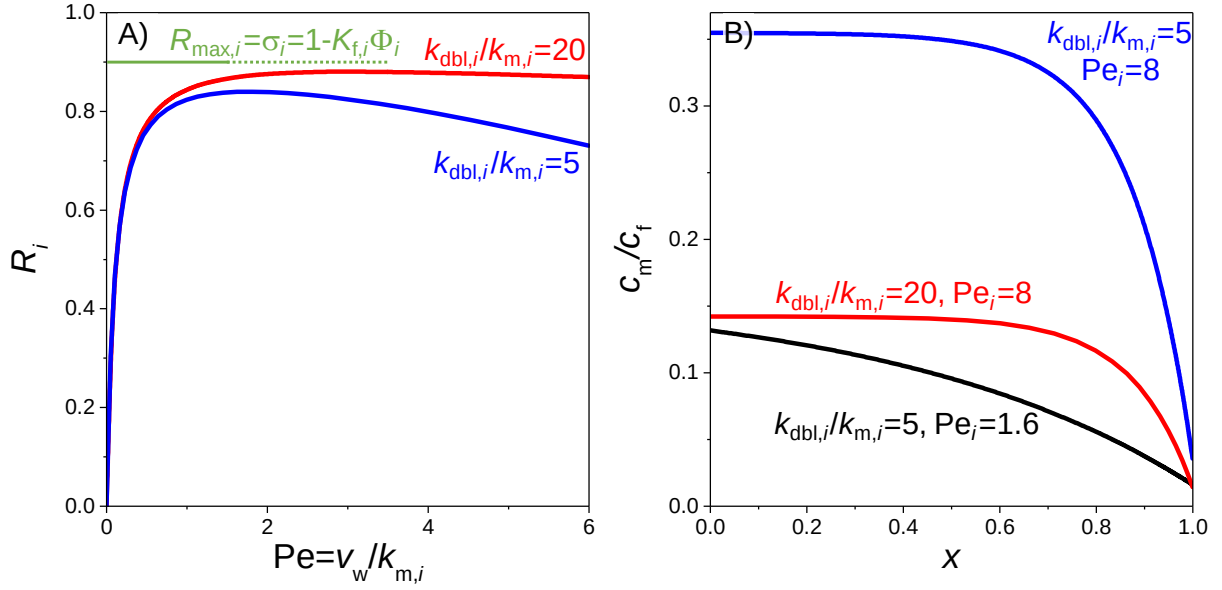


Figure 3: Results of calculations for retention of neutral solutes by an RO membrane in a 1D geometry, combining membrane transport modeling with solute partitioning (corrected SD-model, based on Eqs. (27) and (31)) and concentration polarization (diffusion boundary layer, DBL). A) Retention as function of transmembrane water velocity, v_w , and mass transfer coefficients (with $\sigma_i = 0.9$). B) Examples of solute concentration profiles across the membrane.

membrane), this thickness goes down and $k_{dbl,i}$ goes up.

When we combine all these equations, we obtain the following generalized RO solute retention equation that includes the effect of concentration polarization (Eq. (13) in ref. [4])

$$R_i = 1 - \left[1 + \left((1 - \sigma_i)^{-1} - 1 \right) \cdot \exp(-Pe_{dbl,i}) (1 - \exp(-Pe_i)) \right]^{-1}. \quad (33)$$

When $Pe_{dbl,i} \rightarrow 0$, i.e., when the mass transfer coefficient of the DBL is large enough, for instance because of sufficient stirring, then Eq. (33) simplifies to Eq. (27). Eq. (33) predicts a maximum in retention, as also shown in Fig. 3A, and this maximum retention is at a membrane Pe -number given by $Pe_i = \ln(1 + w)$ where $w = k_{dbl,i}/k_{m,i}$ (Eq. (15) in ref. [4]). The maximum retention is then given by $R_{\max,i} = 1 - \left[1 + \left((1 - \sigma_i)^{-1} - 1 \right) \cdot w \cdot (1 + w)^{-(1+1/w)} \right]^{-1}$ (Eq. (15') in ref. [4]), which for $w \rightarrow \infty$ simplifies to Eq. (31). But in the general case, based on Eq. (33), we can now analytically solve all relevant properties in and near the membrane one by one. Namely, for a known value of v_w , we calculate retention R_i with Eq. (33), then Eq. (2) leads to $c_{p,i}$, J_i follows from Eq. (21), Eq. (32) results in the concentration just on the upstream outside side of the membrane, and then with Eq. (20) we obtain the solute concentration just inside the membrane on that same side. Subsequently we can use Eq. (36), which we will discuss further on, to calculate the pressure that is needed to arrive at the required value of the water velocity v_w . If this velocity v_w is not known, but the applied pressure is known, then an extra iteration is needed, but that is not complicated.

We present in Fig. 3A results for retention as described by Eq. (33) for various values of the ratio of $k_{dbl,i}/k_{m,i}$. Also results are presented for the limiting retention given by Eq. (31). For selected conditions, concentration profiles are presented in Fig. 3B. These profiles are quite strongly non-linear with a steeper decay in concentration near the permeate side. They are not straight lines as would be expected in the SD model.

3.3 Pressures and water flows in a 1D model of RO for neutral solutes

Having discussed how solutes are retained by RO membranes, we next describe the relation between the hydrostatic pressure that is applied across an RO membrane, $\Delta P^{\text{h},\infty}$, and the resulting transmembrane water velocity, v_w . The force balance on the water that will be presented in section 5 describes fluid flow through porous media in the presence of solutes that move through that same porous structure. The theory describes how water flows because of gradients in hydrostatic and osmotic pressure [21] (the latter due to gradients in total solute concentration), and is slowed down by friction with the membrane matrix and by friction with solutes. When solutes also have friction with the membrane (resulting in $K_{f,i} < 1$), then the water velocities decreases: with increasing solute-membrane friction, solutes become more of an obstacle for water to flow. In the opposite limit when solutes ‘just flow along’ with the water, without solute-membrane friction, they do not hinder water transport and then this same theory leads for neutral solutes to the classical Darcy equation for fluid flow through a porous medium. At the edges of the membrane, we have jumps both in osmotic pressure and hydrostatic pressure and it is essential to understand these to predict the direction of water flow, especially in the absence of a significant hydrostatic pressure. These topics are addressed in sections 5 and 6.

Interesting are the changes of pressure across the CP-layer, i.e., across the DBL. The osmotic pressure increases through the DBL, because it is a direct function of the local solute concentration, which also increases towards the membrane (because the solutes are rejected by the membrane while water flows through). At the same time, the hydrostatic pressure does not change through the DBL, i.e., the externally applied hydrostatic pressure ‘arrives at’ the membrane unchanged. Only there, at the outer surface of the membrane, are there jumps in the hydrostatic pressure and osmotic pressure upon entering the membrane [22].

The physical phenomena at the outsides of the membrane are highly intriguing, even for neutral solutes. We know there is a jump ‘down’ in solute concentration upon entering the membrane when solutes are excluded from the membrane, described by $\Phi_i < 1$. A first question is then how the osmotic pressure changes across the interface. The correct analysis leads to the result that also the osmotic pressure goes down, from a value Π^∞ just outside the membrane to a value Π^m just inside the membrane, with $\Pi^m < \Pi^\infty$ when $\Phi_i < 1$ for all solutes. For ideal solutions the osmotic pressure (both inside and outside a membrane) is proportional to the solute concentration according to $\Pi = cRT$ (where c is a total concentration, i.e., a summation over all solutes), and thus the jump in osmotic pressure (counted in the membrane relative to outside) is $\Delta\Pi = (c_m - c_\infty)RT$, and if we now include the partitioning equation we arrive at $\Delta\Pi = -c_\infty(1 - \Phi_i)RT$ (assuming that all solutes have the same Φ_i), which is negative. This means the osmotic pressure goes down upon entry into the membrane. The magnitude of this change is proportional to the solute concentration just outside the membrane, and thus $\Delta\Pi$ can be quite significant on the upstream side of a membrane, while on the downstream side there will hardly be any jump in pressure between outside and inside the membrane.

The next step is to find the relationship between this osmotic pressure change at the membrane interface and the hydrostatic pressure change there. The result is very elegant, and follows from analysis of mechanical equilibrium across this interface, namely that the *total* pressure is invariant across each of the membrane interfaces (between just outside and just inside) [23, 24], with this total pressure a summation of hydrostatic pressure and *minus* the osmotic pressure. This will be further explained in Section 5 as the consequence of the force balance acting on the water evaluated over a very thin layer with rapid changes in pressures, similar to how electrical double layer theory follows from chemical equilibrium across an interface with rapid changes in voltage. Thus, at each

of the two membrane interfaces we have

$$\Pi^m - \Pi^\infty = P^{h,m} - P^{h,\infty} \quad (34)$$

where ‘ ∞ ’ refers to just outside the membrane, and ‘ m ’ to just inside. Thus, a step downward in osmotic pressure upon entry into the membrane (in case $\Phi_i < 1$) translates into an equally large step downward in hydrostatic pressure (Eq. (60a) in ref. [18]). This step downward is the most significant on the side of the membrane with a large solute concentration. [Note that in the SD-model it is stated that only on the downstream side of the membrane there is a pressure drop, with no pressure change on the upstream side of the membrane, see discussion and references in ref. [14]. These ad-hoc assumptions are in contradiction to an analysis based on mechanical equilibrium.]

Inside the membrane water velocity follows from a friction balance that we explain in Section 5, which in a general form is Eq. (90). For neutral solutes, this becomes

$$v_w = -\mathbb{k}_{F-m}^{*,\dagger} \left(\frac{\partial P^{h,m}}{\partial x} - (1 - K_{f,i}) \frac{\partial c_m}{\partial x} \right) \quad (35)$$

where c_m is total solute concentration in the membrane. We assume here for all solutes the same $K_{f,i}$. Here $\mathbb{k}_{F-m}^{*,\dagger}$ is a water-membrane permeability that also includes friction of water with the membrane via a friction of solutes with the membrane. This latter effect disappears when $K_{f,i} = 1$ and then the $\dagger$ -index can be dropped. In that case we will have a linear pressure gradient in the membrane that leads to transmembrane water flow (Eq. (65) in ref. [18]). We can integrate over the membrane thickness, L_m , implement Eq. (34) for mechanical equilibrium, as well as the relation for solute partitioning, and then obtain the classical result that the transmembrane water flow rate is given by

$$v_w = +k_{F-m}^{*,\dagger} \left(\Delta P^{h,\infty} - \sigma_i \cdot \Delta \Pi^\infty \right) \quad (36)$$

where Δ ’s here and in the remainder of this section refer to a value upstream minus that downstream. Eq. (36) is Eq. (10.21) in ref. [25], and is Eq. (41) in ref. [22]. (Note that in the SD-model, see Eq. (12) in ref. [13], the term σ_i is not present.) Interestingly, Eq. (36) describes that for water to flow, the applied pressure must compensate the osmotic pressure *difference* across the membrane (not overcome ‘the’ osmotic pressure at the DBL/membrane interface), and the less so for lower σ_i . Thus if solutes are not very much excluded by the membrane, water will still flow while the hydrostatic pressure can be much lower than the upstream osmotic pressure. Note that ∞ here refers to just outside the membrane, so on the upstream side ∞ refers to a position ‘between’ DBL and membrane. The osmotic pressure at this point is different from that in the bulk phase that is somewhat further away, but the hydrostatic pressure P^h is the same here as further away from the membrane.

In Fig. 4A, we analyze the required transmembrane pressure for any value of the water velocity, v_w , and we notice that for each velocity there is one pressure. It is not the case that we first need to overcome some discrete value of the osmotic pressure of the feedwater before water starts to flow across the membrane. This is because the osmotic pressure that must be overcome is the *difference* in osmotic pressure across the membrane, and retention is low at low water velocities, and then the osmotic pressure on the permeate side approaches that on the feed side, i.e., in the limit of very low transmembrane water velocities, the osmotic pressure difference goes to zero. However, as also shown in Fig. 4A, if the membrane becomes more selective, i.e., the reflection coefficient increases from $\sigma_i = 0.90$ to $\sigma_i = 0.98$, we obtain a curve more in alignment with the common understanding that a minimum pressure is needed to overcome the osmotic pressure,

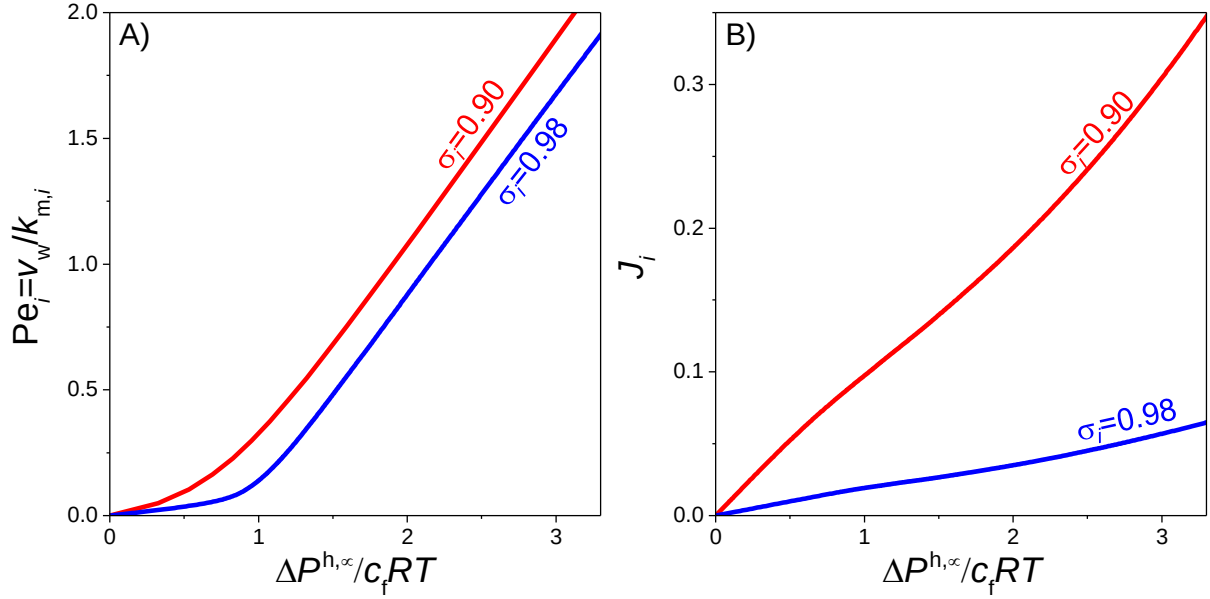


Figure 4: Transmembrane water velocity v_w (expressed as a Pe-number) and neutral solute flux J_i as function of $\Delta P^{h,\infty}$, normalized by the feed osmotic pressure, $c_f RT$, across an RO membrane in a 1D geometry. Results of calculations related to Fig. 3 ($k_{dbl,i}/k_{m,i} = 5$, $\beta = 1$).

before water starts to flow. Furthermore, Fig. 4B illustrates that in the SF model solute flow more or less monotonically increases with pressure. This is very different from what the SD-model predicts, that solute flow is independent of pressure [15].

3.4 Energy efficiency of RO with neutral solutes in 1D geometry

Having analyzed the flow of solutes and water as function of pressure, we can next analyze energy efficiency, η_{RO} , for which results are presented in Fig. 5. To calculate the energy efficiency, we must divide the ‘thermodynamic’ energy, given by Eq. (17), by the real input energy, which is the pressure difference $\Delta P^{h,\infty}$ times the flow of water across the membrane, $\phi_{v,prod}$ (which is equal to the water flowrate v_w times membrane area). When we take the ratio of these two energies, water velocity cancels out, and we obtain $\eta_{RO} = E_{min}/\Delta P^{h,\infty}$. This efficiency η_{RO} starts at zero at very low water velocities (for low Pe_i), and then increases. In the limit of a very high water-membrane permeability ($\beta \rightarrow \infty$, with $\beta = k_{F-m}^{*,\dagger} \Pi_f / k_{m,i}$, where the osmotic pressure of the feed stream is $\Pi_f = c_f RT$), and in the absence of concentration polarization (CP), we can reach an energy efficiency of around 80% when $\sigma_i = 0.90$, which is a maximum value that cannot be further increased at that value of σ_i . When the CP layer plays a more important role, we need a higher pressure at the DBL/membrane interface, to overcome the higher osmotic pressure, and this reduces η_{RO} . And when the water-membrane permeability decreases, efficiency also drops, for instance to values around 40-50% at $\beta = 10$, see Fig. 5. Interestingly, this parameter β does not depend on membrane thickness, so the equations do not simply predict that we have a higher energy efficiency if we reduce membrane thickness. The only effect of thickness is via $k_{m,i}$ which is part of the definition of Pe_i . A thinner membrane leads to lower Pe_i , and can help to increase efficiency (see curve for $\beta = 10$ when we go from high Pe_i back to $Pe_i \sim 0.4$), but for Pe_i already small, a thinner membrane only reduces efficiency because the membrane becomes more leaky and retention goes down which reduces E_{min} and that reduces η_{RO} . This illustrates that –even in this relatively simple example of RO of a neutral solute in a 1D geometry for low water recovery– the effects of the various parameters is not always straightforward. An extended calculation for a full module that we discuss further on will

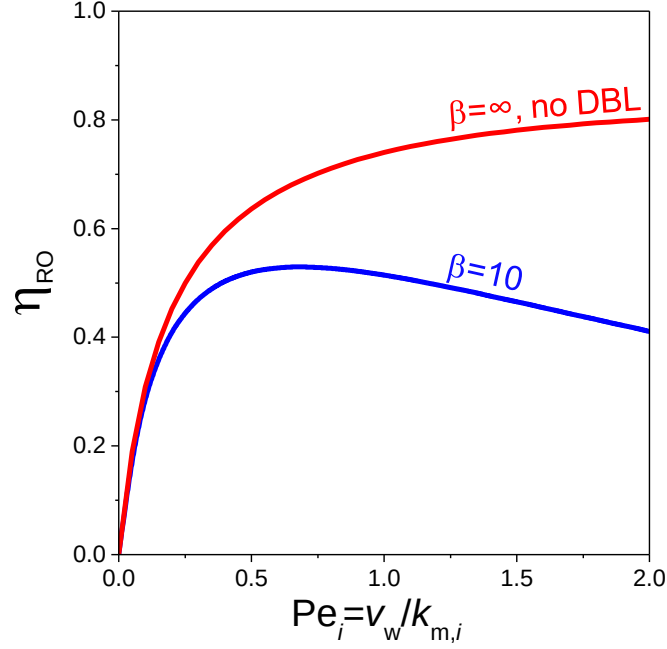


Figure 5: Energy efficiency, η_{RO} , of an RO membrane process with neutral solutes in a 1D geometry. Results of calculations related to Figs. 3 and 4 ($k_{dbl,i}/k_{m,i}=5$, $\sigma_i=0.90$).

demonstrate that the trade-off between water flowrate, freshwater salt concentration, and energy efficiency can be even more complicated.

This analysis finalizes the discussion of RO with neutral solutes in a 1D geometry (limit of low WR). Note that in the final part, relating to pressure and energy, only the simple situation was considered that there is one type of neutral solute. We also treated $k_{F-m}^{*,\dagger}$ as a constant but it is actually a function of the solute concentrations on the two sides of the membrane, and only becomes the constant k_{F-m}^* when $K_{f,i}=1$. Thus, the more complex case that $K_{f,i} \neq 1$ must still be investigated in detail, even for a single solution of neutral solutes. In the next section we analyze a full RO module calculation, again for neutral solutes.

3.5 2D model for an RO module for neutral solutes

We proceed with the analysis of an RO module for the removal of neutral solutes. The model uses analytical expressions for solute and water flow across the membrane at any position in the module, while solving mass balances in the direction between inlet and exit of a flow channel, i.e., in the z -direction along the membrane. So we call this a two-dimensional (2D) model, because that is the physical geometry that is described. The two coordinate axes z on on each side of the membrane are parallel, and we assume the water to flow co-currently, i.e., the exits of both channels are on the same end of the module. On each side of the membrane we assume one value for the pressure, i.e., we neglect pressure changes through the flow channels, that is, along the membrane. Thus, in the entire module there is one value of $\Delta P^{h,\infty}$. On the permeate side there is no inlet, but all water that passes the membrane, going from feed to permeate, leaves as the permeate product stream.¹ We now introduce a dimensionless coordinate axis along the membrane, ξ , which starts at $\xi=0$ at the inlet (feed) where $z=0$, and runs to $\xi=1$ at the outlet of the upstream compartment (retentate).

We use the model for removal of neutral solutes in an RO membrane as discussed in prior

¹The term retentate, with index 'r', from now on refers to the exit of the upstream side. We use 'p' for the z -position dependent permeate concentration within the module, while the exit on this side is denoted as 'p,exit'.

subsections, and we neglect a possible diffusion boundary layer (DBL), and thus we can use Eq. (25) for solute flow, and Eq. (36) for water flow, each evaluated at all positions z (or ξ) in the module. Flow along the membrane is by advection only, thus backdiffusion in the z -direction through the module is not considered. An overall balance can be set up describing that the volumetric inflow of feedwater is equal to the sum of volume flow rates on the upstream side (high pressure side) plus that on the downstream side (low pressure side) at that same z -position. A similar overall balance in salt flow rates also holds. Combination of these two balances leads to

$$c_p = \frac{c_f \phi_{v,f} - c \phi_v}{\phi_{v,f} - \phi_v} = \frac{c_f - c \cdot f}{1 - f} \quad (37)$$

which relates the solute concentration on the downstream side, i.e., the permeate concentration, c_p , at any z -position, to concentrations and flow rates on the upstream side (at the same z -position), and to the feed concentration, c_f . We also introduce here a dimensionless upstream flowrate f , given by $f = \phi_v / \phi_{v,f}$. This factor f starts at $f = 1$ at $\xi = 0$, and decreases with ξ , to reach $f|_{\xi=1} = 1 - \text{WR}$ at $\xi = 1$, according to $f|_{\xi=1} = 1 - \text{WR}$. The z -position-dependent concentration and flowrate on the upstream side are denoted as c and ϕ_v , and they have no additional index from now on.

In steady-state, we have on the upstream side a differential balance for the volumetric flow rate given by

$$\frac{\partial \phi_v}{\partial \xi} = -v_w A \quad (38)$$

where A is the total membrane area in the module. A differential balance for the solute flow rate is

$$\frac{\partial (c \phi_v)}{\partial \xi} = -J_i A. \quad (39)$$

Both these equations can be discretized in various ways, and we use the backward Euler-method, which is equivalent to modeling N ideally stirred tanks placed in series [26]. We can solve these two differential balances together with the appropriate flux equations, such as Eq. (25) for J_i .

Interestingly, these calculations showed that for relevant RO operation the transmembrane velocity, v_w , has values where $\text{Pe}_i \gg 1$, which implies that diffusion across the membrane is negligible, and only advection is of importance, and thus the solute flux can be described by Eq. (30). Intriguingly, if Eq. (30) can be used, then together with Eqs. (38) and (39) we obtain the elegant result that at each position in the module water flowrate and solute concentration on the upstream side are related by

$$\frac{c}{c_f} = \left(\frac{1}{f} \right)^{\sigma_i}. \quad (40)$$

We can apply this equation to the exit of the upstream side of the module where $f = 1 - \text{WR}$ and $c = c_r$. With Eq. (37) we can subsequently calculate the exit permeate concentration, $c_{p,\text{exit}}$.

Indeed, with the development of Eq. (40), we can now quickly evaluate the retention that can be achieved in a module. In the 1D calculation earlier on in this section, in the advection-dominated limit, retention was given by $R_i = \sigma_i$, but in a 2D module, retention is always less, as we will illustrate. We derive an equation for the retention R_i , from combining Eqs. (2), (37), and (40), resulting in

$$R_i = 1 - \frac{c_{p,\text{exit}}}{c_f} = 1 - \frac{1 - (1 - \text{WR})^{1 - \sigma_i}}{\text{WR}} \quad (41)$$

which in the limit of low water recovery simplifies to

$$R_i = \sigma_i - 1/2 \cdot \sigma_i \cdot (1 - \sigma_i) \cdot \text{WR} + \dots \quad (42)$$

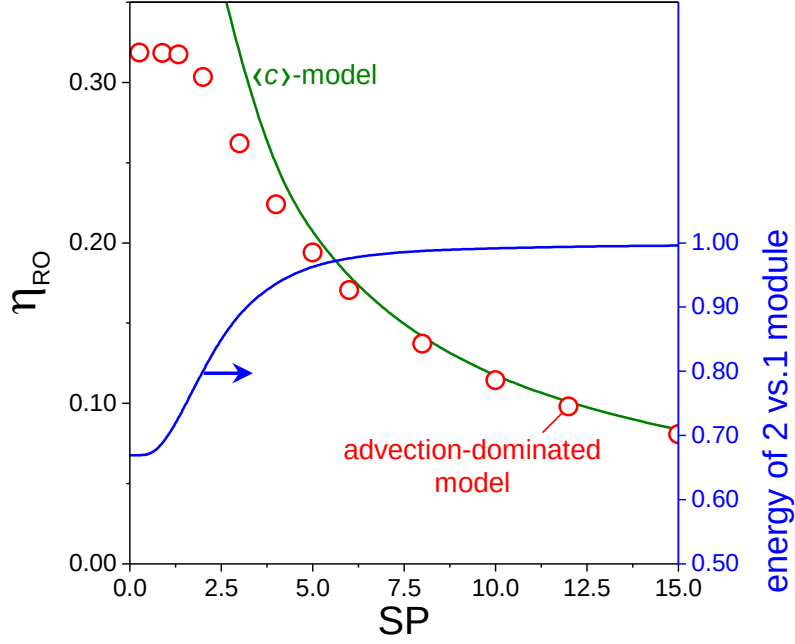


Figure 6: Energy efficiency, η_{RO} , against specific productivity, SP, in a 2D RO module calculation for neutral solutes ($\sigma_i = 0.95$, setpoint for water recovery is $WR = 80\%$ and for retention is $R_i = 90.3\%$ ($\eta_{ERD} = 50\%$). Red circles are from the advection-dominated model and green curve is the analytical $\langle c \rangle$ -model. The blue line describes the energy consumption when using two smaller RO modules versus one module.

which indicates how in the limit of $WR \rightarrow 0$ we end up at the result arrived at earlier for the 1D model, that for very low water recovery we have a retention of $R_i = \sigma_i$, but when WR increases this retention drops. So a high water recovery is nice, but will always go at the expense of retention. This is because of the increasing concentration on the upstream side as we move through the channel, and thus the solute concentration of the water that flows through the membrane further down the module is higher than earlier on, which reduces module retention.

With that part of the model set up, we only need to calculate the applied pressure, which follows implicitly from solving a single differential equation for f versus ξ , which is based on combining Eqs. (36)–(38), leading to

$$-\frac{SP}{WR} \cdot \frac{\partial f}{\partial \xi} = \frac{\Delta P^{h,\infty}}{\Pi_f} - \frac{\sigma_i}{1-f} \left(\frac{1}{f^{\sigma_i}} - 1 \right) \quad (43)$$

where the specific productivity SP is given by $SP = \phi_{v,prod} / (Ak_{F-m}^{*,\dagger} \Pi_f)$, with $\phi_{v,prod} = WR \cdot \phi_{v,f}$ which describes the freshwater production rate per area of membrane. In the numerical calculation, for any value of SP and σ_i , the pressure ratio $\Delta P^{h,\infty} / \Pi_f$ is varied until solving this equation from $\xi = 0$ to $\xi = 1$ leads to f going from $f = 1$ to $1 - WR$.

Now, we will analyze the minimum pressure and maximum efficiency in an RO module. Before setting up a full calculation that includes the differential equation (43), let us first analyze the situation of a low transmembrane water velocity, v_w . This will give us a maximum efficiency and lowest pressure. The pressure required in this limit is what is needed to push the water through the membrane to just reach the required concentration, c_r at the exit of the module, in this limit of a very large membrane area (very low SP). Thus –based on Eq. (36)– the minimum pressure is

$$\Delta P^{h,\infty,min} = \sigma_i (c_r - c_{p,exit}) RT. \quad (44)$$

This pressure can be multiplied by a factor $\phi_{v,f}(1 - \eta_{ERD}(1 - WR))$ where η_{ERD} is the efficiency of

the energy recovery device (to recover energy from the pressure of the retentate flow), to obtain a total energy input. The thermodynamic minimum energy follows from Eq. (15) (without the factor $\frac{1}{2}$ because we discuss neutral solutes at the moment), multiplied by the flow rate of all permeate water produced, $\phi_{v,\text{prod}}$. The ratio of this thermodynamic energy over the total energy input, is the efficiency η_{RO} . For the calculation we discuss below, with $\eta_{\text{ERD}} = 0.50$ and certain values for the setpoints for WR and R_i , we obtain a maximum efficiency of about 32%, see Fig. 6. If we assume that the membrane perfectly blocks all solutes ($\sigma_i \rightarrow 1$), we can derive for the maximum efficiency

$$\eta_{\text{RO,max}} = - \left((1 - \text{WR}) \cdot \ln(1 - \text{WR}) \right) / \left(1 - \eta_{\text{ERD}}(1 - \text{WR}) \right). \quad (45)$$

At the same WR as before of 80%, according to Eq. (45) we have a maximum efficiency of 36%, thus the non-unity value of σ_i in the full calculation leads to a moderate lowering of the RO efficiency. Numerical calculations for the advection-dominated regime, all for $\sigma_i = 0.95$, show that we can increase efficiency by running at a lower WR, up to a maximum efficiency of 46% when WR=55%. Higher efficiencies are possible with higher η_{ERD} with a further reduced WR (for $\eta_{\text{ERD}} = 90\%$, a 68% efficiency at WR=32%). Note that these are maximum efficiencies for the limit of a very low feed flowrate.

We just made an analysis for the limit of very low flowrates, i.e., for low values of SP (very left in Fig. 6, $\text{SP} < 2.5$). Now we discuss the case that flowrates are more significant. It turns out that in these situations a simplified model is highly accurate where we assume that on the downstream side we have the same c_p everywhere. This implies with Eq. (21) that J_i/v_w is also everywhere the same, and now with Eq. (30) we also have the same upstream concentration, which we will denote by $\langle c \rangle$. This we call this the $\langle c \rangle$ -model. Then from Eqs. (21) and (30) we obtain

$$c_p = (1 - \sigma_i) \langle c \rangle \quad (46)$$

where both concentrations are independent of position in the module. These two concentrations relate to the osmotic pressure on the two sides, and thus with Eq. (36) we arrive at a relation between transmembrane pressure and flowrate given by

$$\langle v_w \rangle = k_{\text{F-m}}^{*,\dagger} \left(\Delta P^{\text{h},\infty} - \sigma_i^2 \langle c \rangle R T \right) \quad (47)$$

where we introduce the average transmembrane water velocity $\langle v_w \rangle = \phi_{v,\text{prod}}/A$. (For $v_w = 0$, this equation is Eq. (42A) in ref. [22].) The pressure calculated with Eq. (47) is used to calculate η_{RO} as before, of which results are plotted as a green continuous curve in Fig. 6. So we do not need to solve the differential equation for pressure, Eq. (43). Note that $\langle c \rangle$ is only used in calculating the pressure by Eq. (47), but is not used in the various balance equations. We still use c_r in Eqs. (37) and (40). This simplified model fits very well the results of the advection-dominated model at sufficiently high SP, see Fig. 10. Thus it does not work at a low productivity SP, because there it overestimates efficiency. Thus, the $\langle c \rangle$ -model, and thus Eq. (47), can be reliably used as long as the efficiency it predicts is below the value calculated by Eq. (44) which is for the low-SP limit.

Results of these 2D RO module calculations are presented in Fig. 6, which plots energy efficiency η_{RO} versus the specific productivity, $\text{SP} = \phi_{v,\text{prod}} / \left(A k_{\text{F-m}}^{*,\dagger} \Pi_f \right)$, which describes the volume flow of product (permeate) water per unit membrane area A . The curve depends on water recovery and on σ_i , but interestingly, it does not depend on c_f or on the water-membrane friction, $k_{\text{F-m}}^{*,\dagger}$. A higher $k_{\text{F-m}}^{*,\dagger}$ leads to movement along the curve to lower SP, but we stay on the curve. The same for a lower feed concentration, which via $\Pi_f = c_f R T$ has the effect that SP goes up and thus –moving along the curve– η_{RO} goes down. So to remove solutes for given values of WR and retention R_i , (such as WR=80% and $R_i = 90\%$), the energy efficiency decreases when c_f decreases.

3.6 Efficiency and cost analysis for a 2D RO model according to the $\langle c \rangle$ -model

In the simplest model discussed so far, the $\langle c \rangle$ -model, efficiency η_{RO} depends on SP according to the general function $(SP + \text{cnst})^{-1}$, as we will demonstrate below. But in the relevant range this function can be fitted reasonably well with a simple power law $\eta_{RO} \propto SP^{-\gamma}$, where $\gamma \sim 0.9$. Thus, with the theoretical minimum energy $E_{\min}$ proportional to c_f , efficiency proportional to SP to the power $-\gamma$, and SP inversely proportional to c_f , we end up with the result that for the same setpoints, the energy that needs to be invested ($\propto E_{\min}/\eta_{RO}$) only depends on feed concentration to the power $1-\gamma$, which we find is a power of 0.10. Thus, when we have the same R_i and WR, and when the feedwater solute concentration is reduced by a factor of 10, then the input energy is only less by around 20% ($1-1/10^{0.1}$). Note that we are comparing here two situations where the relative reduction in solute concentration (between feedwater and permeate) is the same, and it is not the case that the permeate final concentration is the same in these calculations. Thus, when we remove trace amounts from water by RO, we will likely end up working at a very low efficiency. Nevertheless, that can still be an acceptable outcome: a process does not have to have high efficiency to be relevant and feasible. Thus, our model predicts that, when water recovery is the same, and we use a single module, removal of solutes from a 1 M solution to reach 100 mM (a factor 10 decrease) requires an energy input that is only 20% higher than when water of 100 mM is desalinated to 10 mM (same factor 10 decrease in concentration). Of course, this calculation depends on the empirical factor γ that we estimated to be $\gamma \sim 0.9$ in the relevant range, and for a different value of γ this calculation comes out differently.

Next we analyze the $\langle c \rangle$ -model and derive an expression for η_{RO} as function of water recovery WR, with retention always described by Eq. (41). For instance, for WR=0.80 and $\sigma_i = 0.95$, this retention is $R_i = 0.9034$. To calculate the efficiency, we first rewrite Eq. (47) to

$$\langle v_w \rangle = k_{F-m}^{*,\dagger} \left(\Delta P^{h,\infty} - \sigma_i (\langle c \rangle - c_p) RT \right) = k_{F-m}^{*,\dagger} \left(\Delta P^{h,\infty} - \frac{\sigma_i^2}{1 - \sigma_i} c_f (1 - R_i) RT \right) \quad (48)$$

and then find energy efficiency η_{RO} as

$$\eta_{RO} = \left(SP + \frac{\sigma_i^2}{1 - \sigma_i} (1 - R_i) \right)^{-1} \frac{\ln(c_r/c_f) - WR(1 - R_i) \ln(c_r/c_p)}{1 - \eta_{ERD}(1 - WR)}. \quad (49)$$

For a perfect separation, when $\sigma_i \rightarrow 1$, thus $R_i \rightarrow 1$, $c_p \rightarrow 0$, and $c_r = c_f/(1 - WR)$, then Eq. (49) simplifies to

$$\frac{1}{\eta_{RO}} = \left(\frac{1}{WR} - \frac{SP}{\ln(1 - WR)} \right) \left(1 - \eta_{ERD}(1 - WR) \right). \quad (50)$$

This remarkably simple result is arrived at from Eq. (49) when we implement Eq. (41) to replace $1 - R_i$, and take the limit that for small y , $(1-x)^y \rightarrow 1 + y \cdot \ln(1-x)$. We replace $E_{\min}$ based on Eq. (15) by Eq. (16). Note that if here in Eq. (50) we set $SP = 0$, we do not arrive at Eq. (45), because that equation is based on the advection-dominated model, for the plateau at low SP, and the $\langle c \rangle$ -model on which Eq. (50) is based, is only valid at high SP. Similar to what is depicted in Fig. 6, these equations result in a monotonically decaying η_{RO} when SP increases, i.e., we can work at high efficiency, η_{RO} , but we will then have a low SP. Or we work at high SP but we must then accept a lower efficiency.

Based on these equations for efficiency, we can make a preliminary cost analysis. Here we do that on the basis of Eq. (50). For the more general Eq. (49) this is more complicated because in a module retention depends on water recovery. Optimal RO operation minimizes the sum of two cost factors, first the energy use, and second the material costs. The energy costs are the product of five

factors: 1. the costs of a unit of energy CUE (in an arbitrary monetary unit, per J of energy); 2. a multiplier m_1 for the total energy use in a unit or plant, over that required to generate the pressure in the RO unit; 3. the factor RT ; 4. the feed flow rate $\phi_{v,f}$; and 5. the required energy per m^3 feed stream. The material costs are the product of four factors: 1. the costs to purchase the membrane, CMP (in the same monetary unit as above, per m^2); 2. a multiplier m_2 to relate to other plant expenses, assumed to be proportional to the installed membrane area; 3. the required area A ; and 4. one over the lifetime of the membrane, i.e., the operational time Δt that a membrane can sustain before it must be replaced. The energy per m^3 feed stream is the applied pressure multiplied by a term that accounts for the recovery of pressure energy, thus $\Delta P^{h,\infty}$ times a factor $1 - \eta_{\text{ERD}}(1 - \text{WR})$. We add up energy and material costs to arrive at the total costs (per time of operation)

$$\text{TC} = m_1 \cdot \text{CUE} \cdot RT \cdot \phi_{v,f} \cdot \Delta P^{h,\infty} \cdot (1 - \eta_{\text{ERD}}(1 - \text{WR})) + m_2 \cdot \text{CMP} \cdot A / \Delta t. \quad (51)$$

Because we define efficiency as $\eta_{\text{RO}} = E_{\min} \phi_{v,\text{prod}} / (\phi_{v,f} \cdot \Delta P^{h,\infty} \cdot (1 - \eta_{\text{ERD}}(1 - \text{WR})))$, Eq. (51) can be rewritten to

$$\text{tc} = \frac{\alpha e_{\min}^c}{\eta_{\text{RO}}} + \frac{1}{\text{SP}}, \quad \alpha = \frac{m_1 \cdot \text{CUE} \cdot k_{\text{F-m}}^{*,\dagger} \cdot \Pi_{\text{f}}^2 \cdot \Delta t}{m_2 \cdot \text{CMP}} \quad (52)$$

where $e_{\min}^c = E_{\min} / \Pi_{\text{f}}$, tc is the dimensionless total costs, and where the cost factor α is dimensionless as well. This factor is proportional to the cost of energy, and inversely proportional to the costs of purchasing the membranes. Thus less expensive membranes increase this factor α . In Eq. (52) we can insert Eqs. (16) and (50) and then minimize with respect to SP to find the best operational point, which can then be inserted in Eq. (51) to calculate the costs at that condition. Combination of Eqs. (16), (50), and (52), and minimization of the function tc with respect to SP leads to

$$\text{SP}|_{\min} = \sqrt{\frac{\text{WR}}{\alpha (1 - \eta_{\text{ERD}}(1 - \text{WR}))}} \quad (53)$$

which shows that when energy costs increase, thus α increases, that the system should be operated at higher efficiency and lower SP, i.e., it becomes favourable that more membranes are purchased and installed. And the same result is obtained for the case when the purchase costs of membranes goes down (or we make improvements such that we can use them longer, i.e., Δt goes up), then α also goes up, and we also want to run the desalination plant at lower SP and higher efficiency η_{RO} . Thus, because of cost optimization, ‘less expensive membranes have higher efficiency’. This conclusion, here arrived at for neutral solutes, likely also applies for RO with salt solutions. In the next section we make a similar analysis for ED.

Besides this optimization with respect to SP, also an optimization as function of WR is possible. But this is more complicated because retention is a function of water recovery. So either we must then assume retention to be perfect, or we must include an extra model element such as a bypass of feedwater to one of the exit streams, to keep retention at the same value throughout the optimization. We assume a perfect solute retention. Then we numerically find the operational optimum with respect to both WR and SP on the basis of Eqs. (50) and (52). With an energy recovery device with an efficiency $\eta_{\text{ERD}} = 0.5$ and an initial cost factor $\alpha = 0.01$, then the system has the following optimized properties: $\text{WR} \sim 0.888$, $\eta_{\text{RO}} \sim 0.191$, and $\text{SP} \sim 9.70$. As Fig. 7A shows, when the factor α goes up, either because material costs are lowered or the price of energy goes up, the new optimum is at a higher energy efficiency η_{RO} , with SP going down because we push the water through at a lower rate per unit membrane area, with a significantly increased size of our module: in Fig. 7A, we can see that SP goes down from ~ 9 to ~ 3 when α is increased by a factor 10, which then implies that it becomes favorable to increase the total area in the membrane modules by a factor of ~ 3 (for

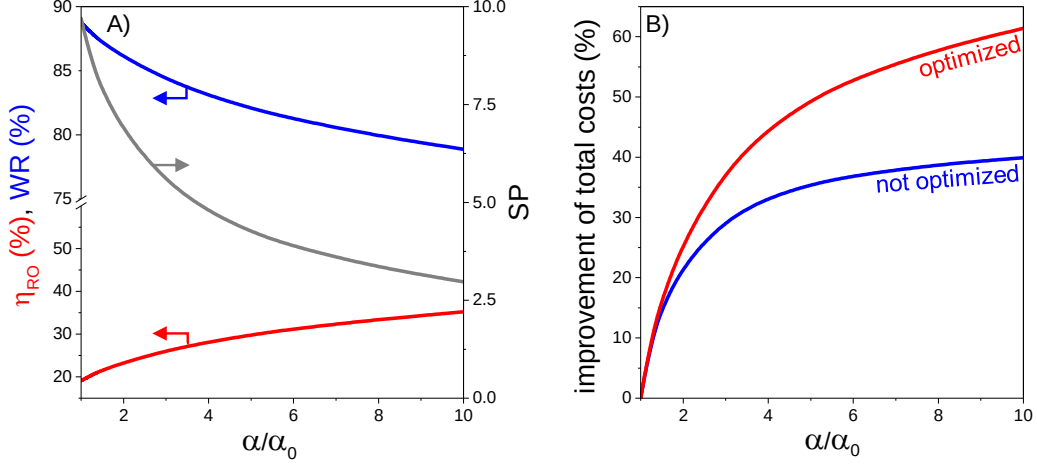


Figure 7: Results of cost optimization of RO module for perfect retention, $R_i = \sigma_i = 1$, as function of an increase in the cost factor α/α_0 , leading at each α to a different optimum for WR and SP, and a different η_{RO} . Panel B is based on α increasing because material costs go down, with energy costs unchanged.

the same amount of water that is treated). The optimal water recovery decreased from around 88% to 78% when α increases by a factor of 10.

We now assume that the increase in α is because we are able to reduce the material costs, either by extending the lifetime of membranes or by reducing CMP, both of which will increase the cost factor α . We then arrive at a percentage-wise improvement (reduction) in total costs as depicted in Fig. 7B, first simply by factoring in that material costs go down, without redoing the optimization, and secondly after a renewed optimization which further improves the cost reduction. The corresponding new values of η_{RO} , WR, and SP, are depicted in Fig. 7A. The end result is that the energy efficiency of the process goes up significantly, and this happens in a sense ‘because’ the materials were less expensive.

Of course this analysis was only an example calculation, and a large range of factors is neglected, such as the effect of the DBL on transport, the dependence of fouling on transmembrane flowrates, and other process constraints. And again, this entire calculation was for a membrane that under all conditions perfectly blocks the solutes, which is not the case in reality. And the calculation was for a neutral solute, not for a salt solution... Another assumption was that the hydrostatic pressure on the upstream side was set to a constant factor, neglecting the decrease in pressure on the upstream side of the module, between feed and retentate (order of 0.2 bar) [27]. Another cost factor that is relevant to include is that of water pretreatment. Pretreatment costs scale linearly with feed flow rate, so is inversely proportional to water recovery (for a given system permeate production). Yet another factor is brine disposal which has an associated cost [27], and if that is proportional to the volume thereof (and not on the total amount of salt in there), then another factor must be included that scales with $(1-WR)/WR$, which would shift the optimum to a higher WR. Optimization studies that incorporate these effects are very useful.

3.7 Optimization for two combined RO modules in a system

Let us next analyze what advantage can be achieved when we work with two RO modules, not one, as also analyzed in ref. [28]. When we compare with a single module, the ratio of total water throughput over total membrane area must be the same, i.e., with two modules, on average they

must be twice as small. We return to the advection-dominated model (not the $\langle c \rangle$ -model). In the calculation we make, the retentate of module 1 is directed to module 2 that operates at a higher pressure. The same overall setpoints apply as in section 3.5. We again use $\eta_{\text{ERD}} = 0.5$.

What we find is that there is a positive effect of the use of multiple units when we work at low SP, in the region where efficiency is at a maximum, see Fig. 6, a limit that was described by the advection-dominated model based on Eq. (43), and was also obtained from the low-SP limit given by Eq. (44). Now, in that limit, with two modules in series, efficiency η_{RO} can increase significantly at low SP, from 0.32 to 0.48. At this optimum, water recovery is almost the same in both units, at a value of ~ 0.55 , similar to statements in ref. [28] and references in there. At this optimum, the pressure in the second unit is about $2.2\times$ that of the first module. (Thus the retentate of module 1, which is about 45% in volume of the feed stream, is further increased in pressure by $1.2\times$ what was the pressure applied to the water entering the first module.) This improvement in efficiency needs the two pressures to be chosen exactly right. If they are not, efficiency can even be lower than when only a single module is used.

However, with increasing SP the advantage decreases and the model predicts that from some value of SP onward, there is no advantage any more, and the use of two modules is even less effective than a single module, also when we allow for the additional option that one module is smaller than the other. From that point onward, the increasing pressurization when entering the second module is not offset by the advantage of working at two pressures. This is then a result opposite to what was found for low SP. That using multiple modules is not always better, may go against common intuition on the design of RO systems where the general idea is that more units is always better, because then we do not have to push so hard in the first module where solute concentrations are still low. However, in that theoretical analysis the total membrane area may inadvertently have been allowed to increase (while the total freshwater production was invariant), which is then not a fair comparison. Instead, we must make such comparisons with the total membrane area the same [27].

3.8 Outlook

In the previous sections, we discussed various aspects of the modeling of an RO unit to remove neutral solutes, based on both 1D and 2D geometries, using models that are more detailed, and others that are more approximate. Several simplifications were made throughout this section. For instance, the 2D calculations never included the diffusion boundary layer (DBL, or CP layer).

It is also important to investigate whether the various results obtained in this section come out the same for desalination, i.e., for the removal of salt. Though we will not address that problem with the same level of detail as we just did for neutral solutes, several key results for a two-dimensional model of an RO module with a symmetric 1:1 salt solution will be discussed in Section 6.

4 A simple model for electrodialysis for symmetrical cell pairs

A simple model for electrodialysis can be derived by an approach that we will outline next. In electrodialysis feedwater flows into two adjacent sets of channels, see Fig. 1B. In one array of channels the water will be desalinated, these are the diluate channels, denoted by ‘d’. The salt molecules move to the concentrate ‘c’ channels. In many practical ED stacks, the flow patterns in each channel is in first approximation ‘at 90 degrees’ to the flow pattern in adjacent channels. This is called a cross-flow geometry. This geometry is somewhat more complicated to model, though

there are semi-analytical methods, and thus in this section we assume that we have perfect co-current flow, i.e., we assume a one-dimensional geometry where the two flows enter the module from the same side and flow in the same way on either side of the membranes. The two channels can still be different in width and the flowrates (thus residence time per channel) can be different.

An ED stack has two electrodes on either end with related ‘electrolyte compartments’ where electronic current is transferred to ionic current. In this review we focus on the repeating unit of an ED stack, which is the membrane cell pair, see Fig. 1B, which consists of two membranes and two flow channels. As mentioned, in the flow channels we will assume co-current flow of the water along the membranes, from inlet to exit of the channel. While the water flows through these thin channels between the membranes, ions move from the d-channels to the c-channels by transport across the membranes. This process is possible because –driven by the electrical current– anions move in one way out of each d-channel, to be transported through membranes that are selective for anions. These are the ‘anion-exchange membranes’ or AEMs, which are membranes in which the water-filled pores are lined with high concentrations of positive membrane charges (of the order of 5 M fixed membrane charge per volume of water in the membrane). The cations move in the other direction, passing cation-exchange membranes, or CEMs. These CEMs have a high concentration of fixed negative charge, again of the order of 5 M, and they preferentially allow access to cations, largely blocking the passage of anions. Each d-channel has such an AEM on one side, and a CEM on the other. The net effect of this layout of a cell pair, of a sequence of an AEM, d-channel, CEM, and c-channel, and this repeated hundreds of times, is that the d-channels are being desalinated, while the salinity in the c-channels increases between start and exit of each channel. Thus in the AEMs the anions are the main charge carrier, while in the CEM cations are the main carrier of charge. These ions in their respective membrane are the counterions (anions in AEM, cations in CEM). The minority ions, that ideally are fully blocked, are called coions (cations in AEM, anions in CEM).

Alternative explanation of ion fluxes and leakage in an ED module. Let us again describe the functioning of the ED cell pair, in a slightly different manner. With the current directed to the right, cations will flow rightward, and anions will flow leftward. In the bulk of the channels, for a symmetric system, this current is equally carried by cations, moving right, and anions that move left. Cations and anions can have velocities in this left-right direction that tend to zero (near a blocking membrane), but they do not flow ‘back’ at any point in the module. With AEMs and CEMs alternately placed in the stack, and with cations moving right, anions left, then the following will happen. One type of channel has a CEM on its right side. That membrane freely allows the cations to pass to the next channel (to the right). And that same channel has an AEM on its left side, and anions can pass that membrane freely and move left. But now, in these adjacent channels, things are different. The anions want to continue moving left, but now they encounter a CEM which blocks them. And cations that want to move right, encounter an AEM that blocks them. So in this second set of channels, the ions accumulate, and thus the salt concentration increases. These channels will be the concentrate channels. Ions that arrive in these channels, cannot escape.

What was just described, is the ideal functioning of an ED stack, and it is generally like this early on in a cell pair, where current densities are still high and concentration differences between the two types of channels, c_d and c_c , still low¹. But later on in the stack, with $c_c \gg c_d$, then *ions start to leak from the concentrate channel*. They leak in the direction that drives them onward, *not back to the diluate channel that they came from*, but on to the *next* diluate channel. The membrane that blocked them from leaving the concentrate channel in the ideal case (e.g., the AEM for a cation), becomes leaky. Why is that? Chemically the membrane is of course the same as before. But the increased coion leakage from the concentrate channels is because the increasing concentration difference across this membrane leads to more and more diffusion of coions. Indeed, diffusion is the main membrane transport mechanism

for co-ions in ED. Thus further on in a cell pair, the driving force for diffusion increases, and thus ions that were kept in the concentrate channel quite well at first, now increasingly leak out; so cations that entered a concentrate channel on the left side, now leak out on the right side.

Now it turns out to be the case that this leakage is more pronounced for membranes that have a high diffusion coefficient, and for membranes that are thin. With thin membranes, the gradient in concentration across the membrane is steep, and we therefore have a large driving force for diffusion [29]. To avoid this leakage from the concentrate channel, membranes should not be too thin (a thickness of at least 10 μm is a well-known estimate). This increased leakage is a reason why, towards the exit of a channel, the net rate of salt removal goes down (per unit membrane area). The other reason is that the local current density goes down, so the transport of ions from diluate to concentrate channels (the intended purpose of ED) also goes down. This decrease is because the resistance goes up in the diluate channel, and Donnan potentials increase. So for a given cell pair voltage, V_{cp} , the high current densities that could be achieved early on in the channel, are dropping significantly the further one progresses through the channels. Thus the flux of cations and anions arriving in the concentrate channels goes down significantly. And at the same time diffusional leakage goes up, as discussed. As a consequence, at some point in the channel a condition is reached that current is still running, but the net salt transport from diluate to concentrate channels (arrival minus leakage) drops to zero. At this point (and beyond), the *current efficiency*, λ , reaches the value of zero. This limiting value of $\lambda = 0$ will be used further on to calculate the ‘final’ current density and salt concentrations, c_d and c_c , in a stack run at a fixed cell pair voltage.

¹ The same analysis applies to a (short, or ‘differential’) stack that is fed from reservoirs to which the effluent also flows back (batch-mode ED operation, solution is recirculated between the reservoirs and the stack, and is desalinated over time). Then statements about ‘early on in the stack’ can be changed to ‘early in time’, and ‘further on’ by ‘later’, assuming we started with two reservoirs at equal salt concentration. Note that in this layout, the stack can be run at constant voltage or constant current. Instead, in the calculation of co-current flow in a single stack without these reservoirs, then in the common design with two undivided end-electrodes, then the entire stack is operated at one value of the cell pair voltage, and the further we are away from the entrance, the lower is the current density.

In a later section we discuss how to optimize ED operation by having multiple ED stacks in series with different water velocities and operational currents, but for now we discuss a single cell pair. In the simple model we discuss for now, we neglect water transport through the membranes. Unfortunately, in reality this water flow is not zero, and has a tendency to lower ED performance, but for now we neglect it. Thus the water recovery, WR, is directly set by how much of the feedwater flows to the c-channels and how much to the d-channels, namely $WR = 1/(1 + \phi_{v,c}/\phi_{v,d})$. Thus, for equal flow rates to the two channels, we have $WR=0.50$.

We will set up for each channel a steady-state plug flow model, with ion transport through the membranes in a ‘sideways’ direction. In each channel, at each position z , where z is parallel to the membranes, there are concentration profiles across the channel, from membrane to membrane, but they can be relatively minor when the channels are narrow. [This is not always the case, and sometimes these profiles do matter significantly. Also with three or more different ions, instead of a simple binary salt, the situation in the flow channels can become radically different, and concentration profiles across the channel must then be considered.] Thus for the moment in each channel we use a z -dependent mass balance that does not have any dependence on position in the perpendicular x -direction, which is the direction towards the membranes, at right angles to the z -direction of water flow along the membranes. Then a mass balance for an ion i at some position in a flow channel becomes

$$p \frac{\partial c_i}{\partial t} = -v_z \frac{\partial c_i}{\partial z} \pm a (J_{AEM,i} + J_{CEM,i}) \quad (54)$$

where p is the porosity (open volume fraction) of the channel, v_z the velocity of the water through

the channel, in the direction along the membrane. This is an average velocity, i.e., the water flowrate in m^3/s divided by the cross-sectional area of a channel (the cross-section through which the water flows). It is a superficial velocity, i.e., the presence of a spacer material that reduces p does not change the numerical value of v_z . The specific membrane surface area is a , which is the area of one membrane lining a channel divided by the channel volume. And thus $a = 1/L_{\text{ch}}$, where L_{ch} is the width of the channel. We use the symbol $\pm$ in Eq. (54) to avoid technicalities of how to define the directions of fluxes, because that is not essential at this moment.

We now consider a binary 1:1 salt. We add up Eq. (54) for anions and cations, assume steady-state, thus the term $\partial c_i / \partial t$ is zero, and at each position z , we have electroneutrality, i.e., $c_+ = c_-$, and from now onward we use c_j for the salt concentration, i.e., $c_j = c_{+,j} = c_{-,j}$ which is z -dependent, and different between the c- and the d-channels (which are referred to by subscript j). Thus we add up Eq. (54) for the two ions, and then divide by 2. The summation of the four fluxes $J_{i,\text{IEM}}$ is twice the total salt flux, J_{salt} , which is the flux of salt from d- to c-channels, per unit area of one membrane, as a summation of the two membranes together. We then arrive at

$$v_z \frac{\partial c_j}{\partial z} = \pm a_j J_{\text{salt}} = \pm a_j \lambda_{\text{cp}} \frac{I}{F} \quad (55)$$

where we introduce the current efficiency λ_{cp} , a process parameter that is z -dependent in the cell pair, i.e., $\lambda_{\text{cp}}(z)$, and we introduce the current density I in A/m^2 divided by Faraday's number, F . In an ED stack, in a cell pair, from entrance to exit (i.e., from $z = 0$ to $z = \ell$, with ℓ the length of the channel in flow direction), current density I will change with z . We cannot make the current constant in a normal ED stack. This is because there are two end-electrodes that have a certain voltage between them. Therefore it is the cell pair voltage, V_{cp} , which will be the same at each z -position in the cell, and because Donnan potentials and resistances increase with z , current will decrease in that direction.

Eq. (55) introduces a cell-pair based current efficiency λ_{cp} which relates to the total salt transport flux between the d- and c-channels, J_{salt} , and the current, I , all z -dependent. To calculate λ_{cp} and J_{salt} we must solve at each z -coordinate a membrane model for the AEM and the CEM, and in each membrane describe the flow of counterions and coions. In such a model all parameters, including the diffusion coefficients of the ions, and the membrane charge, can be different between AEM and CEM. This calculation is possible, but in this review we simplify this situation, and assume the AEM and CEM are each other's perfect 'mirror image'. Thus, the fixed membrane density in an AEM, denoted by the symbol X , has the same magnitude as the membrane charge of a CEM, only the sign of these two charges is opposite to one another. Furthermore, the diffusion coefficient of the counterions in the AEM and CEM is the same, and the same for the coions (coions and counterions can still have different diffusion coefficients). With these assumptions, the cell-pair based λ_{cp} is now the same as a 'single membrane'-based current efficiency λ_{th} , where we use the subscript 'th' for 'theoretical' because this single membrane-efficiency cannot be measured but can be obtained from a theoretical model of a single ion-exchange membrane (IEM), as we will do in this section.

Thus, in each of the two channels of an ED cell pair, we now have the mass balance

$$v_z \frac{\partial c_j}{\partial z} = \pm a_j \lambda_{\text{th}} \frac{I}{F}. \quad (56)$$

Eq. (56) can be solved as function of z if we know $I(z)$ and $\lambda_{\text{th}}(z)$, and to do that, at each position z two algebraic equations (AEs) must be solved jointly with Eq. (56) in each channel. Because the right side of Eq. (56) is the same for each channel, except for the $\pm$ sign, and except for a_j , we can

add up these two balances over the two channels, and arrive at

$$L_{\text{ch}} v_z \frac{\partial c}{\partial z} \Big|_{\text{d-channel}} + L_{\text{ch}} v_z \frac{\partial c}{\partial z} \Big|_{\text{c-channel}} = 0 \quad (57)$$

where we implemented how in each channel $a_j = 1/L_{\text{ch},j}$. Eq. (57) can be integrated from the inlet to a certain position z . When we have an equal inlet concentration $c|_{z=0}$, namely equal to the feedwater concentration, c_f , we arrive at

$$\phi_{v,d}(c_d(z) - c_f) + \phi_{v,c}(c_c(z) - c_f) = 0 \quad (58)$$

where we converted from flow velocities v_z to flow rates in each channel, $\phi_{v,j}$, by multiplying each side by the geometrical ‘depth’, which is the membrane area divided by the path length ℓ . Thus we end up with an equation describing how all salt that is removed from one channel ends up in the other channel. Eq. (58) is an AE that becomes part of the model and it replaces the ODE for one of the channels. Thus, the full ED model (co-current, steady-state, symmetry, ...) consists solely of one ODE and three AEs, to be ‘tracked’ along a coordinate z from entrance to exit of the cell pair. Note that still WR can have any value, and also the width of each channel can be different, thus the residence time in the d- and c-channels can be very different.

Now, interestingly, for any c_d we see from Eq. (58) that c_c automatically follows, and as we show below, for a given cell pair voltage, V_{cp} , then λ_{th} and I also follow automatically. Thus, there is no particular dependence on position z . Thus, we can also rewrite the one remaining ODE, that we will evaluate for the d-channel, and we take I as a positive number, to

$$\frac{\partial c_d}{\partial t^*} = -a_d \lambda_{\text{th}} \frac{I}{F} \quad (59)$$

where the time-on-stream, t^* , is given by $t^* = z/v_z$. This representation shows that this model is equally valid for two very different cases: first, the above-discussed geometry of an ED cell pair with two co-current flow channels, but the same model also applies for an ED stack with relatively low desalination ‘per pass’ where the effluents of each channel are rerouted to two storage tanks from which the same channel is fed again. Now t^* really is a time. The area/volume ratio, a_d , in Eq. (59) is changed to the area of one membrane times the number of cell pairs, divided by the volume of the d-reservoir (including volumes of tubing and the channels in the ED stack). The final modification is that in Eq. (58), we replace volume flow rates by the two reservoir volumes. Note that the model as presented assumes no volume flow through the membranes, but this effect can be included with few modifications. Let us point out again that despite the appearance of a ‘time’ t^* , nevertheless at the level of the membrane this is a steady-state process model.

To solve this model, we can calculate the time step Δt^* when we let c_d step downward in fixed increments starting at an initial value. We can use the Crank-Nicolson scheme in which λ_{th} and I are averaged between known values at subsequent ‘ c_d -lines’, see Ch. 21 in ref. [26], at k and $k-1$, according to

$$\frac{c_{d,k} - c_{d,k-1}}{t_k^* - t_{k-1}^*} = -\frac{a_d}{F} \frac{1}{2} \left(\lambda_{\text{th},k} I_k + \lambda_{\text{th},k-1} I_{k-1} \right). \quad (60)$$

Now we must only formulate expressions for current efficiency, λ_{th} , and for current density, I , as function of c_d and cell pair voltage V_{cp} . This is discussed in the next section.

4.1 Calculation of current and current efficiency in ED

To calculate current and current efficiency, we first must relate current density I to the voltage drops over the membranes and flow channels. Indeed, we can relate the cell pair voltage, V_{cp} , to

the voltage drops across two channels (d- and c-channels), two membranes (AEM and CEM), and in total four Donnan potential drops (at each channel/membrane interface in the cell pair). For a symmetric system, the four Donnan potentials simplify to two times the same difference between two Donnan potentials. In the symmetric model we also assume each membrane has the same resistance. The two channels will, as desalination progresses, have very different resistances. In each membrane, the relationship between current and voltage can be described by

$$I = \pm k_m |X| F \Delta \phi_m \quad (61)$$

where k_m is a membrane transport coefficient, $|X|$ is the magnitude of the membrane charge density, and $\Delta \phi_m$ is a dimensionless potential, or voltage, drop across the membrane. This voltage drop is only across the inner coordinates of the membrane, excluding the voltage changes (Donnan potential) at the two edges of each membrane. These we will discuss further on. Eq. (61) assumes that both ions have the same diffusion coefficient in the membrane, D_m , and thus the mass transfer coefficient k_m , which is the ratio of this D_m over membrane thickness, L_m , is the same for both ions. This diffusion coefficient will be much lower than in free solution, because of membrane porosity, tortuosity, and the small pore size. For instance the diffusion in the membrane will be lower by a factor of 10–50 compared to outside, in free solution. Eq. (61) furthermore assumes that the membrane charge density, $|X|$, is large compared to salt concentrations outside the membrane. (In a generalized version of Eq. (61), the term $|X|$ is replaced by an average total ions concentration, $\langle c_{T,m} \rangle$.) Any voltage ϕ can always be multiplied by the thermal voltage, $V_T = RT/F \sim 25.6$ mV (at room temperature), to arrive at a dimensional voltage in (m)V.

Similar to Eq. (61), we have a current-voltage relationship across the d- and c-channel, given by

$$I = \pm 2k_{ch} c_j F \Delta \phi_{ch,j} \quad (62)$$

where k_{ch} is a channel transfer coefficient, given by $k_{ch} = \varepsilon D_\infty / L_{ch}$, where D_∞ is the diffusion coefficient of ions in free solution. The factor ε is the channel porosity p divided by a tortuosity factor τ , i.e., $\varepsilon = p/\tau$, and describes how in the spacer channel the effective diffusion coefficient for electromigration is lower than in free solution, because of a spacer structure that fills the channel. The channel width is L_{ch} . Note the factor 2 in Eq. (62) which relates to both anions and cations contributing equally to current flow. Eq. (62) is valid when the two ions have the same diffusion coefficient,² and in the absence of concentration gradients across the channel.

Next we consider the four Donnan potential drops at the four membrane/solution interfaces. In the symmetric system we discuss, the two Donnan drops at the AEM are exact mirror-images of those at the CEM because we assume the same $|X|$ for AEM and CEM, so we only have to discuss one of these membranes here. We continue to assume that we only have monovalent ions in solution. We neglect all ion partitioning effects other than the Boltzmann distribution that relates to the charge of ions. This Boltzmann equation for each ion, distributing between just in the membrane and just outside, is

$$c_{m,i} = c_{\infty,i} \exp(-z_i \Delta \phi_D) \quad (63)$$

where ∞ refers to a position just outside the membrane (in the ED model z -dependent), and Eq. (63) applies to the interface of the membrane with electrolyte in the d- and c-channels. The Donnan potential at each of these interfaces is the potential just in the membrane, relative to just outside, in solution.

²As long as we have a binary salt, also for ions of different valency and diffusion coefficient, an expression such as Eq. (62) can always be arrived at, but with an appropriately averaged diffusion coefficient.

Based on electroneutrality in the membrane, valid at each position in the membrane, we have

$$c_{m,+} - c_{m,-} + \omega|X| = 0 \quad (64)$$

where ω is the sign of the membrane charge ($\omega = +1$ for an AEM and $\omega = -1$ for a CEM). Based on Eqs. (63) and (64) we can derive for the Donnan potential that

$$\Delta\phi_D = \sinh^{-1} \frac{\omega|X|}{2c_\infty} \quad (65)$$

where the function $\sinh(x)$ is identical to $\frac{1}{2}(\exp(x) - \exp(-x))$, and $\sinh^{-1}$ is the inverse of that function (also written as asinh or $\operatorname{arcsinh}$). Eq. (65) is valid for a solution with monovalent ions only, and without considering partitioning coefficients other than pure ‘Boltzmann’ electrostatics. See refs. [10] and [26] for a detailed discussion on the various contributions to the partitioning coefficient at the interface between free solution and a microporous charged material. As Eq. (65) shows, for a membrane with negative fixed charge, i.e., a CEM, which needs to ‘pull in the cations’, the Donnan potential is negative, i.e., from just outside to just inside the membrane, the potential goes down.

The eight voltage drops described in the above equations can all be added up (taking care of appropriate choices of $\pm$ -signs), to arrive after multiplication with V_T , at the cell pair voltage, V_{cp} . And this V_{cp} is invariant with z . [Thus the eight individual voltages described above, they all change with z , but their summation, which is V_{cp} , doesn’t.]

Before we show the result of such a combined expression for V_{cp} versus current, I , and versus channel salt concentrations, c_d and c_c , we evaluate the Donnan potential, Eq. (65), in a novel, useful, manner. The Donnan potential, as described by Eq. (65), must be evaluated on two sides of the same membrane (and then for two membranes), and because both count ‘inside minus outside’, the result for a given membrane is

$$\Delta\phi_{D,tot} = \sinh^{-1} \frac{|X|}{2c_d} - \sinh^{-1} \frac{|X|}{2c_c} \quad (66)$$

which we can call the total Donnan potential across a membrane. [This excludes the membrane potential across the inner coordinate of the membrane due to current, as given by Eq. (61)]. We drop here and below the subscript ∞ when we refer to salt concentrations just outside the membrane. We next evaluate the equation for $\omega = -1$, i.e., a CEM, but for an AEM the same result is obtained with a final overall minus-sign difference. Now, generally, this equation is not used in this way, perhaps because it is considered too complicated. Instead, in literature the following equation can often be found, which is

$$\pm \Delta\phi_{D,tot} = \ln \frac{c_c}{c_d} \quad (67)$$

and this expression is valid for an ideal membrane, one that perfectly blocks coions. There is no dependence on membrane charge here. A prefactor in front of the $\ln$ -term can be added, which is then called permselectivity, which describes the difference between the ideal $\ln$ -term, and the measured Donnan potential, a number that is below unity. The $\pm$ -symbol depends on the sign of the membrane charge and on whether ‘ Δ ’ is defined from c-side to d-side or vice-versa.

Now, interestingly, Eq. (67) follows from Eq. (66) as a first term in a Taylor-expansion around the point that $1/X \rightarrow 0$. But this first term is only very accurate when on both sides of the membrane the salt concentration is very low compared to X . But we can now add the second term in the Taylor expansion and obtain a novel, highly accurate, expression for the Donnan potential, given by

$$\pm \Delta\phi_{D,tot} = \ln \frac{c_c}{c_d} - \frac{c_c^2 - c_d^2}{X^2} \quad (68)$$

where the total Donnan potential decreases (in magnitude) when membrane charge density X drops, or the difference between c_c and c_d increases. Note that this novel equation is only valid as a first correction to Eq. (67) when X is at least of the order of the salt concentrations. It cannot be used for even lower X , because at very small X , Eq. (68) becomes unrealistic, and we should add more terms in the Taylor expansion. Note that in Eqs. (66) and (68) it is important to use the same unit for c and for X , either both in M or both in mM, i.e., mol/m³.

Thus, the new expression provides a correction to the Donnan potential for finite values of X , without invoking an empirical ‘permselectivity factor’. The additional term correctly identifies that the correction to ideality depends on the membrane charge density X , but also on c_d and c_c , and this clearly highlights how the permselectivity factor, that was historically used, is not a membrane property, but is a term that for a given membrane varies with salt concentration, even along the z -coordinate within the same ED stack (or changes in time in a batch experiment with changing reservoir concentration).

In general, for arbitrary salt mixtures, and with other partitioning effects besides pure electrostatics, it is advisable to return to a Boltzmann equation for each ion, Eq. (63), in combination with electroneutrality in the membrane. In extension, we can include how the membrane charge is a function of pH and concentration of other ions, such as the Ca²⁺-concentration just in the membrane (because of a protonation reaction, or ion adsorption), and these ion concentrations in turn depend on the Donnan potential, thus on X . Also other acid-base associations between ions must be included, such as the protonation of an ion, to go from a basic to acidic state, such as NH₃ associating with an H⁺-ion to NH₄⁺, the distribution of which depends on pK and local pH [30]. Furthermore, mass transport effects across the channels (especially for solutions with three or more types of ions) will lead to ion concentrations just outside the membrane (which are the concentrations that are used in the Donnan equilibria) that are different from the channel-averaged concentrations at that z -position. Also transmembrane water flow can be of importance, because it ‘pushes’ ions away from the membrane surface, or pulls them in, and thus also modifies ion concentrations just next to the membrane. All in all, in detailed modelling of an ED cell, the above simplified equations must always be critically evaluated to ascertain whether they are sufficiently accurate for the situation under study.

Now we can finally proceed to adding up all these eight voltages to obtain one final equation. The result shown next assumes the c - and d -channel to have the same width, L_{ch} , and we already discussed the assumption that each membrane has the same charge X and the same transport properties, thus the same k_m . We then arrive at

$$\frac{V_{cp}}{V_T} - 2 \left(\ln \frac{c_c}{c_d} - \frac{c_c^2 - c_d^2}{X^2} \right) = \frac{I}{F} \left(\frac{1}{2k_{ch}} \left(\frac{1}{c_d} + \frac{1}{c_c} \right) + \frac{2}{k_m|X|} \right) \quad (69)$$

where c_c , c_d and current density I are all z -dependent. The placements of the factors ‘2’ at the end of this equation may seem counterintuitive, with one in the numerator, one in the denominator, but this is correct. We can use Eq. (69) to derive the current at the very start of the stack (or, cell pair), because then both c_c and c_d are equal to c_f , and thus all Donnan effects disappear, and we arrive at

$$V_{cp} = \frac{RT}{F^2} \left(\frac{1}{k_{ch}c_f} + \frac{2}{k_m|X|} \right) I \quad (70)$$

which includes two (actually four) resistances in series at each z -position in an ED stack, namely the resistance in each channel and in each membrane. Each resistance is the inverse of a conductivity, and these are respectively $2k_{ch}c_f$ and $k_m|X|$ (both with units mol/m²/s).

Finally we evaluate the current efficiency λ_{th} , at each z -position. To that end we must establish what is the flux of counterions through each membrane, which is the desired effect, leading to

desalination, and the flux of coions, which is a deleterious effect, reducing desalination, because coions go in the wrong way, from the c-channels to the d-channels. Based on fluxes through a single membrane, we define λ_{th} as

$$\lambda_{\text{th}} = F \frac{J_{\text{m,ions}}}{I} = \frac{J_{\text{m,+}} + J_{\text{m,-}}}{J_{\text{m,+}} - J_{\text{m,-}}} \quad (71)$$

where $J_{\text{m,ions}}$ is the total ions flow through a membrane, $J_{\text{m,ions}} = J_{\text{m,+}} + J_{\text{m,-}}$, and the current density is $I = F (J_{\text{m,+}} - J_{\text{m,-}})$. The factor λ_{th} is only defined for desalination of a 1:1 salt, and will always be between 0 and 1, with $\lambda_{\text{th}} = 1$ for ideal operation, with only counterion transport through each membrane. In the limit of $\lambda_{\text{th}} = 1$, for each unit of charge a full salt molecule is removed from a diluate channel, while in the limit of $\lambda_{\text{th}} = 0$ there is current running through the membrane, but there is no net salt removal. This last situation develops near the exit of an ED cell where all ions arriving in a concentrate channel move out again to a next diluate channel, see a box on p. 27. The ions leak out of the c-channel because the concentration is so much larger here than in the d-channel, and because the current is low because of a high resistance in the diluate channel, and because of the high Donnan potentials near the exit of an ED cell.

Next, we analyze ion transport, both of counterions and coions, across a membrane. To that end we use the Nernst-Planck (NP) equation that can be solved at any position in the membrane together with local electroneutrality. For each ion, the NP-equation is

$$J_i = -D_{\text{m},i} \left(\frac{\partial c_{\text{m},i}}{\partial x} + z_i c_{\text{m},i} \frac{\partial \phi}{\partial x} \right) \quad (72)$$

where ϕ is the inner-membrane potential, and x is a coordinate directed across the membrane (across the shortest distance, i.e., across the thickness). We neglect advection in the membrane (water flow and the drag thereof on ions). We assume the same diffusion coefficient for anions and cations in the membrane, and thus we can first subtract anion flux from cation flux, to arrive at

$$I/F = J_+ - J_- = -D_{\text{m}} c_{\text{T,m}} \frac{\partial \phi}{\partial x} \quad (73)$$

where $c_{\text{T,m}}$ is the total concentration of the two ions together in the membrane, i.e., $c_{\text{T,m}} = c_{\text{m,+}} + c_{\text{m,-}}$. Let us reiterate again, that all equations in this section only apply to a 1:1 salt with only one type of cation and one type of anion. Even adding a second cation with lower diffusion coefficient than the first, can significantly change the model, and even more so when this other ion also has a different valency.

Note that current density I in Eq. (73), calculated based on ion transport in a membrane, *is the same* current density that also crosses (at that z -position) each of the flow channels, and crosses the other membrane too; i.e., there is (at each z -position) only one value of current I , the same at each location across each of the channels, and it has the same value in all the other cell pairs (at that z -position), which are all exact copies of one another. Related to this statement is that this model assumes that all current ‘lines’ cross the stack at right angles to the channels and membranes, i.e., the current crosses these layers in the shortest direction. This is a highly accurate assumption. With regards to current maldistribution in ED, the only point that must be carefully addressed is that in ED it is possible that current leaks through the electrolyte circuit of redox-active ions that flow along the two electrodes to facilitate the conversion between ionic and electronic current. The tubes that transport this ‘electrode solution’ must have a much higher resistance than the pathway for current to cross all membranes and channels in an ED stack. Thus the electrode solution circuit must use long and thin tubing. Another option is to electrically ‘open’ the electrical pathway in this circulation by the use of a drip system.

The NP equation for anions and cations also leads to an expression for the total ions flux

$$J_{\text{m,ions}} = J_+ + J_- = -D_m \left(\frac{\partial c_{\text{T,m}}}{\partial x} - X \frac{\partial \phi}{\partial x} \right). \quad (74)$$

In the derivation of Eq. (73) and (74), the electroneutrality condition in the membrane is implemented. Note that Eq. (74) makes use of X without a $|\dots|$ -statement, and indeed X here can be negative and positive. Because of steady-state both equations can be integrated across the membrane, leading to

$$I = \pm k_m F \langle c_{\text{T,m}} \rangle \Delta \phi_m \quad (75)$$

where the term $\langle c_{\text{T,m}} \rangle$ is based on an appropriate average of $c_{\text{T,m}}$ between the two sides of the membrane (from just in the membrane very near the diluate channel, to very near the concentrate channel). It is a highly accurate approximation to take the straight average of the two values of $c_{\text{T,m}}$ at the two sides of the membrane (still just inside the membrane), i.e., $\langle c_{\text{T,m}} \rangle = \frac{1}{2} (c_{\text{T,m}}|_d + c_{\text{T,m}}|_c)$.

How to obtain this total concentration $c_{\text{T,m}}$ at each side? To that end, we return to the earlier Donnan analysis. For a 1:1 salt and no additional partitioning effects, we arrive at

$$c_{\text{T,m},j} = \sqrt{X^2 + 4c_{\infty,j}^2} = |X| + 2c_{\infty,j}^2 / |X| + \dots \quad (76)$$

i.e., this total ions concentration is independent of the sign of X (see full expression), and is always larger than $|X|$, i.e., always $c_{\text{T,m},j} \gg |X|$. Now, for a highly charged membrane, with $|X|$ much larger than the two salt concentrations on both sides, it is approximately correct to assume that $c_{\text{T,m},j} \sim |X|$. That approximation was used in the derivation of Eq. (61) and thus in the final result, Eq. (69), but officially an appropriate average of $c_{\text{T,m}}$ must be used there instead of $|X|$.

The integration of Eq. (74) across the membrane, possible because $J_{\text{m,ions}}$ is independent of position in the membrane, results in

$$J_{\text{m,ions}} = \pm k_m (\Delta c_{\text{T,m}} - X \Delta \phi_m) \quad (77)$$

without any further approximation. We calculate λ_{th} according to Eq. (71) by implementing Eqs. (73) and (77), and obtain

$$\lambda_{\text{th}} = 1 - k_m F \Delta c_{\text{T,m}} / |I| \quad (78)$$

where we made the replacement $\langle c_{\text{T,m}} \rangle \rightarrow |X|$.

At high enough $|X|$ (relative to c_c and c_d), from Eq. (76) we can derive that the difference in $c_{\text{T,m}}$ across the membrane can be simplified to $\Delta c_{\text{T,m}} = 2(c_c^2 - c_d^2) / |X|$, a result that we can implement in Eq. (78) to arrive at

$$\lambda_{\text{th}} = 1 - \frac{2k_m F}{|X||I|} (c_c^2 - c_d^2) \quad (79)$$

which is the final equation that we need to solve the full co-current ED model. Let it be reminded that all parameters such as fluxes and potentials in Eq. (77) and other equations for membrane transport all depend on position z in the module (or alternatively, depend on time-on-stream, t^*).

Eq. (79) is interesting in its own right, as it shows that λ_{th} will be high, near unity, at high currents and a high membrane charge, and for low salt concentration differences, but λ_{th} will go down for lower currents and lower membrane charge. It also goes down when k_m goes up, which happens when the membrane thickness is reduced. Indeed, for very thin membranes, ED does not work, because then there is a large flux of coions across the membranes that opposes desalination [29]. [The same problem is found in ‘reverse electrodialysis’ with membranes that are too thin or too permeable or not charged high enough: there is a large salt flow across the membranes, so the potentially large power is not realized because the salt solutions in the two

channels are rapidly mixed up.] Eq. (79) illustrates that λ_{th} will depend strongly on position in the cell, because current I will change with z , and concentrations c_d and c_c will do the same. So current efficiency is not a fixed number, it is not a property of the membrane.

Calculation results are made with this model for co-current flow in ED for a constant V_{cp} across the cell pair (the same value of V_{cp} at each position z). Results show that while the d-channel is desalinating, the Donnan potential strongly increases on the diluate side (by a value which exceeds the decrease on the concentrate side), the current goes down, and current efficiency decreases, finally to drop all the way to zero, i.e., $\lim_{z \rightarrow \infty} \lambda_{th} = 0$, see Fig. 8. We can use this result to calculate from Eq. (79) the final salt concentration in a single-pass co-current ED cell pair when the cell is large enough, together with Eqs. (58) and (69). These three equations can be solved analytically when we start with a certain final diluate concentration, c_d , then calculate c_c , then $|I|$, and finally V_{cp} . Calculation results are presented in Fig. 9A and show the final, i.e., minimum, c_d that can be reached in an ED stack as function of cell pair voltage V_{cp} for two values of water recovery, WR. Also the current in this ‘steady-state’ limit is plotted. In all cases the stack is fed with artificial seawater, i.e., a 0.5 M 1:1 salt solution. Other parameters that we use are $X = 4.0$ M, $k_m = 1$ $\mu\text{m/s}$, $k_{ch} = 5$ $\mu\text{m/s}$, and $L_{ch} = 200$ μm .

The calculation just discussed describes how deep we can potentially desalinate as function of cell pair voltage, and the final, ‘steady state’, current. But it does not give us the average current, which we need to calculate the electrical energy input. To that end we analyze the full model and for a certain residence time t^* and specific area $a_d = 1/L_{ch}$, we calculate as function of V_{cp} the exit value for c_d , and the average current density $\langle I \rangle$ (in A/m^2). This calculation, of which results are presented in Fig. 9B, also shows a result familiar in the ED literature, that with increasing $\langle I \rangle$, at some point V_{cp} diverges. This divergence is generally attributed to concentration polarization (CP), which in the context of ED is the *decrease* of salt concentration towards the membranes in the diluate channels. Interestingly, we also find the same divergence in the present model that does not include the CP effect yet. In the present model, the voltage diverges when we reach complete desalination, because the resistance in the diluate channel then rapidly increases as well as the Donnan potentials at the diluate-membrane interfaces. We will discuss a more extensive 3D ED model including concentration polarization in Section 6.4.

In summary, the co-current ED model is a powerful and insightful model to discuss and quantify several well-known observations in ED. In the next section we use the model to relate stack energy efficiency η_{ED} to the specific productivity of freshwater in ED, a relation that we need in a full cost optimization.

4.2 Energy efficiency and cost optimization of an ED stack

With an ED model available, we can now address the question how to decide at what current to run an ED stack, and how much membrane area to use. To that end we must balance energy and material costs, in a fashion similar to what we did for RO in Section 3.6. We again set up a relationship between energy efficiency, now given by the symbol η_{ED} , and the volume flow rate of product water per unit membrane area, expressed as a specific productivity, SP, see Fig. 10. In ED electric energy is the main input of energy, given by the product of cell pair voltage and average current density, and thus energy efficiency is given by

$$\eta_{ED} = \frac{\phi_{v,d} E_{min}}{\langle I \rangle A V_{cp}}. \quad (80)$$

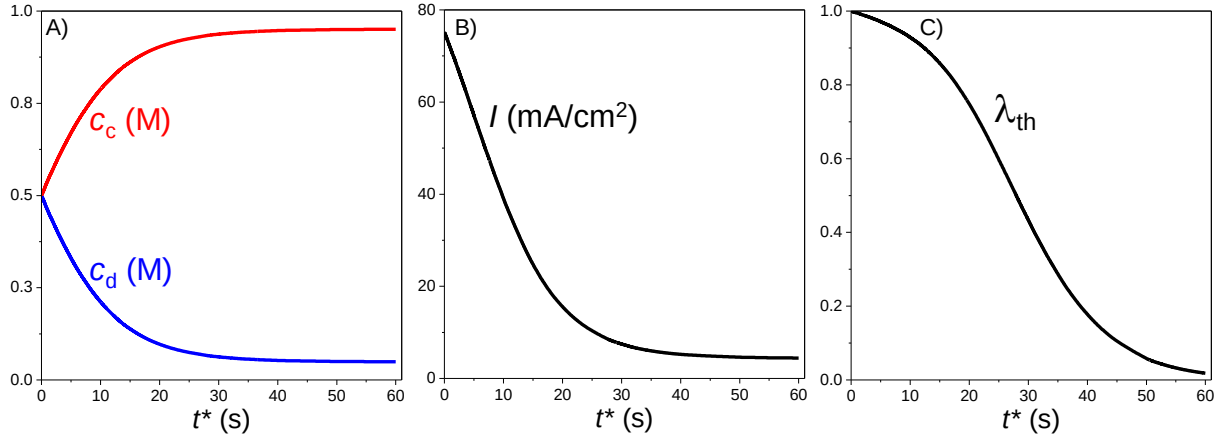


Figure 8: Calculations of co-current flow in electrodialysis with a diluate and concentrate channel for $V_{cp} = 180$ mV, $c_f = 0.5$ M, $WR = 0.5$, and $X = 4.0$ M. As function of time-on-stream, t^* , the salt concentrations in the two channels change, first rapidly, then slowly, while current I goes down as well as current efficiency, λ_{th} . In steady-state (at large t^*), current still runs, but there is no more salt transport, thus $\lambda_{th} \rightarrow 0$.

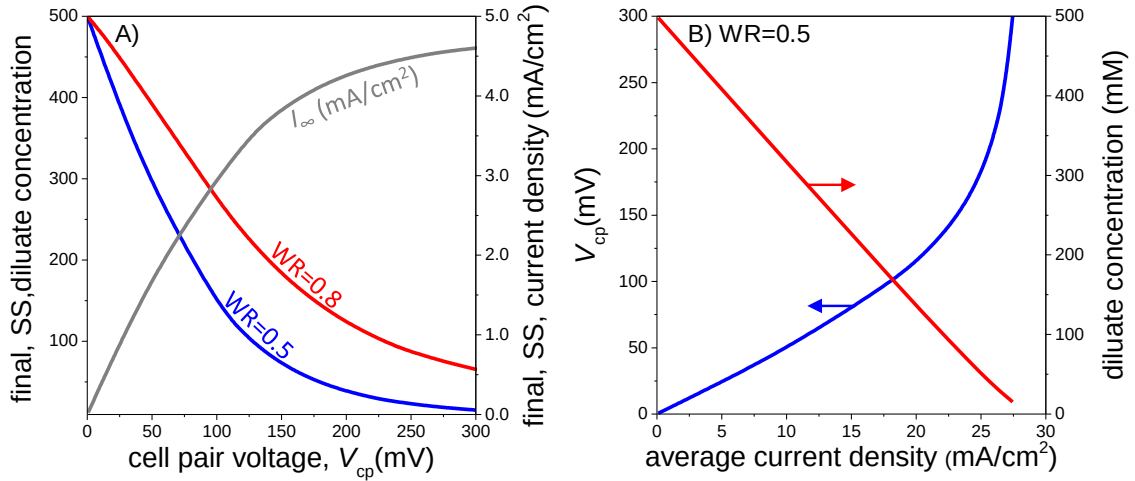


Figure 9: Analysis of concentrations and current density using the co-current ED model. A) For two values of WR we analyze as function of V_{cp} the ‘steady-state’ diluate concentration, c_d , which can be reached after a long enough residence time t^* , as well as the resulting final current density for $WR=0.5$. B) For a fixed $t^* = 40$ s ($SP=5$), we vary the average current density $\langle I \rangle$ and find that V_{cp} diverges above a certain current, when desalination goes to completion.

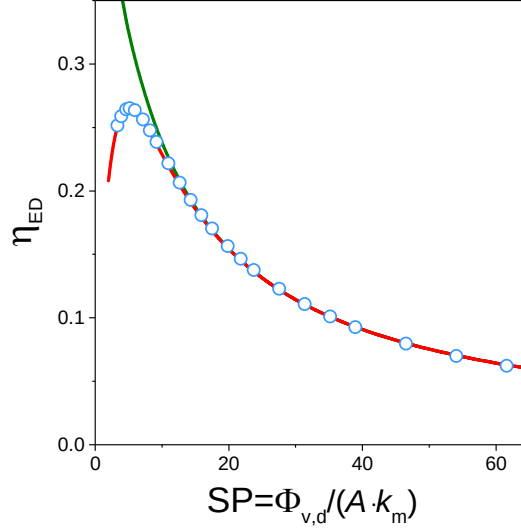


Figure 10: Energy efficiency in ED, η_{ED} , versus specific productivity, SP, according to the co-current ED model for the situation analyzed in Figs. 8 and 9 (blue circles). In all calculations a 500 mM salt solution is desalinated to 50 mM, and water recovery is 50%. Fit line according to $\eta \propto (SP + \text{cnst})^{-1}$ (green) and more accurate fit (red line) can be used in cost optimization studies.

Just as for RO, also for ED we set up an equation for the total costs per time of operation, TC, and this equation is basically the same as Eq. (51) [31]

$$TC_{ED} = m_1 \text{CUE} \langle I \rangle AV_{cp} + m_2 \sum_j \text{CMP}_j A_j / \Delta t_j \quad (81)$$

which includes the costs of electrical energy, and next a summation over AEMs and CEMs of their purchase costs per m^2 , their area, A_j , and their lifetime, Δt_j . We now assume that all three of these factors are the same for the two membranes. We insert Eq. (80), divide by $\phi_{v,d}$, multiply by the membrane transfer coefficient, k_m , and obtain

$$tc_{ED} = \frac{\alpha_{ED}}{\eta_{ED}} + \frac{1}{SP}, \quad \alpha_{ED} = \frac{m_1 \cdot \text{CUE} \cdot E_{\min} \cdot \Delta t_j \cdot k_m}{2m_2 \cdot \text{CMP}} \quad (82)$$

where $SP = \phi_{v,d} / (A k_m)$, with A the area of one membrane in the cell pair. We can minimize η_{ED} if we know its dependence on SP, for a given value of α_{ED} . Based on the co-current ED model, we can generate a list of calculation outputs for a specific membrane system, see the ‘data points’ in Fig. 10, and fit with an equation that subsequently is used in the cost optimization study. Interestingly, the same as for RO, see Fig. 6, we obtain a curve of η vs. SP with a maximum efficiency at low SP, and a decrease in η at lower and higher SP. In the present case, the orange fit line in Fig. 10 has the same structure as that which followed from the $\langle c \rangle$ -model for RO, namely $\eta \propto (SP + \text{cnst})^{-1}$. A more accurate fit line (green) also provides a correct description of efficiency at very low SP, by ‘multiplying the red line’ with an empirical function, $\tanh(SP^{0.8}/3.2)$. Note that these functions do not need to have a physical basis; they are just fit functions that ‘summarize’ real calculation results in a ‘continuous’ function that we need in a cost optimization study. When the purchase costs of membranes goes down, this minimization shows that we should run at lower SP, i.e., for a given production of freshwater we should install more membranes, and run them at a higher efficiency. Thus one can quip that ‘cheaper membranes are more efficient’ just as was the case for RO.

5 Fundamentals of ion and water transport across membranes

5.1 Introduction

In the previous sections we made use of simplified approaches for the transport of solutes and water through RO and ED membranes. In the RO-calculation we neglected charge effects, while in the ED-model we neglected ion-membrane friction and advection of ions. In a generalized approach all these driving forces and frictions are jointly considered, in both types of processes. Ongoing acid-base reactions between ions, often also involving the H^+ -ion, is of crucial importance as well to include. On both outsides of the membranes there is equilibrium (across the interface) both with regards to pressure (mechanical equilibrium), and with regard to chemical potential of solutes (chemical equilibrium). These equilibria also depend on non-electrostatic contributions to a partitioning of solutes across the water-membrane interface because of affinity and volumetric, ‘excess’, effects, or perhaps because of other physical forces [4]. For transport across the membrane, be it for membranes as thin as 100 nm as in RO, and membranes 100 to 1000 times thicker for ED, we can start with continuum theory from the field of electrokinetics of thin channels, and apply this theory to transport in the pores in RO and ED membranes. These models, such as the space charge (SC) model [32–34], then become much more simple because profiles in concentration and potential over the cross-section of a pore in the membrane then do not need to be considered. This simplified ‘uniform potential’ (UP) model, i.e., the SC-model without including the radial direction of the pore, describes ion transport and water flow through the pores of a membrane, and considers the effect of pressure on both membrane interfaces [35–38]. The UP-model is thus an extension of the Teorell-Meyer-Sievers (TMS) model that describes ion transport across membranes, but TMS theory does not include water flow. The UP-model can now be extended to incorporate many more effects than we could in the SC model. Besides water-membrane and water-ion friction, we can now also include ion-membrane friction, and also ion-ion frictions, the latter describing how ions that overtake slower ones drag them along, and vice-versa how the slow ions retard fast ones. In the two-fluid model (TFM) [39, 40] this detailed model that includes various frictional forces is extended with a description of water flow [41, 42].

In the two-fluid model, a local force balance acting on solutes (such as ions) includes driving forces and frictions. Driving force acting on a solute are mathematically formulated as a gradient of an ion’s chemical potential. Frictions experienced by a solute are with the water flowing through the pores, and with the membrane, which behaves as an immobile background structure. For the water a similar force balance is set up but driving forces acting on the water are hydrostatic and osmotic pressures gradients [25]. Thus ions and other solutes are described in a very different manner from how the water that flows along all ions through the pores is described. This is one of the key points of the two-fluid model (TFM). This is different from Maxwell-Stefan (MS) theory that was successfully developed for multicomponent gas flows in porous media where all gas substances can be described in like fashion. This MS theory has also been applied to electrolytes, i.e., the flow of ions and water through porous media, but it then faces several shortcomings. MS theory applied to electrolyte flow relies too heavily on a statistical-mechanical view wherein the molecular nature of water is included and all ions and other solutes just like each water molecule occupy one of the available statistical sites. This electrolyte version of MS theory depends too much on water flowing according to its own concentration gradient, as if it is also a dispersed species, which it is not. Instead, water is a continuum fluid that fills all space between the discrete, dispersed entities which are the ions and other solutes. MS theory applied to electrolytes does not consider the difference between total, hydrostatic and osmotic pressure, and generally has a hard time implementing partitioning and equilibrium conditions on the outer surfaces of membranes. Instead, all of these

effects are comprehensively included in the two-fluid model, which describes the flow of ions and other solutes through tortuous media simultaneous with solvent flow, using different expressions for both. To illustrate how well TFM works, we show further on a precise description by TFM of the phenomenon of osmosis in an experiment where water and solutes flow in opposite directions through an ion-exchange membrane, a problem that is very hard to describe precisely. But first we present the basics of flow of ions (and other solutes) and water in TFM.

5.2 Flow of solutes through membranes in the two-fluid model (TFM)

To describe the flux of ions through a porous tortuous medium, we must first of all set up an expression for the chemical potential of an ion. Ions can be hydrated to some extent, and the water molecules that are firmly bound in its water shell are part of what we call the ion. In this section the porous medium is a membrane, so at some point the explanation switches from ‘porous medium’ to ‘membrane.’ We can consider multiple contributions to the chemical potential of an ion (unit J/mol) in a porous medium, adding up to

$$\mu_i = \mu_{\text{ref},i} + RT(\ln c_i + z_i \phi) + \mu_{\text{aff},i} + \mu_{\text{exc},i} + v_i P^{\text{tot}} \dots \quad (83)$$

where the reference value $\mu_{\text{ref},i}$ is of importance when ions are part of chemical reactions, where $\ln c_i$ relates to diffusion (ion entropy), $z_i \phi$ to electrostatic effects, $\mu_{\text{aff},i}$ to an affinity, leading to (one contribution to) a solubility S_i or partitioning coefficient Φ_i , and where finally $\mu_{\text{exc},i}$ is a function that describes various volumetric interactions between ions, and of ions with the porous medium [10]. The pressure insertion term $v_i P^{\text{tot}}$ relates to the molar volume of an ion v_i and the total pressure P^{tot} equals a hydrostatic pressure, P^h , minus osmotic pressure, Π .

Across any thin interface, where we can assume mechanical equilibrium, the total pressure is invariant, as we will discuss in Section 5.3, so in a partitioning function, as below, this term cancels out. For the volumetric term, $\mu_{\text{exc},i}$, when only interactions of an ion with the porous medium are of importance (and not between ions), then it is a constant factor that can then be grouped with $\mu_{\text{aff},i}$. Then, for a neutral solute, we arrive for the distribution of a solute across a porous medium/solution interface (between ‘in’ and ‘out’), at

$$\frac{c_{i,\text{in}}}{c_{i,\text{out}}} = \exp\left(-(\mu_{\text{aff},i} + \mu_{\text{exc},i})/RT\right) = S_i = \Phi_i \quad (84)$$

where the solubility S_i or partitioning coefficient Φ_i is a constant if the excess term does not depend (too much) on concentrations just outside and inside the pore. Also other effects can contribute to Φ_i when they lead to an additional thermodynamic penalty (or attraction) for an ion to enter a porous medium. The excess contribution is often described as function of an ion size-pore size ratio, λ , described for a perfectly cylindrical pore and spherical ions to $\Phi_i = (1 - \lambda_i)^2$, but many more accurate expressions are available, see Ch. 4 in ref. [26]. For ions, with a charge different from zero, we can extend Eq. (84) with a Boltzmann (Donnan) term, and we then end up with

$$c_{i,\text{in}} = c_{i,\text{out}} \Phi_i \exp(-z_i \Delta \phi_D) \quad (85)$$

which extends Eq. (63), by now including a partitioning effect that has a non-electrostatic origin. It must be realized that now ion concentrations are defined per unit open pore volume. Otherwise, when defined per unit total volume of the porous medium (including both pores and membrane structure), then the excess term must be considered to explicitly include a porosity effect $-\ln(1-p)$, which in the end leads to the same result. So it is easier to immediately describe how concentrations are per unit open pore volume in the porous medium (membrane) and then we can then leave out this $-\ln(1-p)$ -term.

In the two-fluid model, Eq. (83) is used to describe the driving forces acting on an ion as minus the gradient of chemical potential. And these driving forces are compensated by all frictional forces, according to a force balance given by

$$\mathcal{F}_{\text{driving},i} + \mathcal{F}_{\text{friction},i} = 0. \quad (86)$$

Assuming for simplicity ions not to have volume –but for a full description including effects of ion volume, see Ch. 7 in ref. [26]–, thus $v_i = 0$, we obtain for the driving force acting on a mole of ions

$$\mathcal{F}_{\text{driving},i} = -\frac{\partial \mu_i}{\partial x} = -RT \left(\frac{1}{c_i} \frac{\partial c_i}{\partial x} + z_i \frac{\partial \phi}{\partial x} \right) \quad (87)$$

where we consider a single coordinate, x . An extension to multiple coordinates or dimensions is straightforward. We can leave out here a gradient-term related to $\mu_{\text{aff},i}$ because this ion-membrane affinity will be invariant across the membrane. We also leave out the excess-term that relates to volume effects. Interestingly, even when assuming $v_i = 0$, it is not automatically the case that this term is zero, because interactions with other solutes that do have volume can lead to a non-zero $\mu_{\text{exc},i}$. But nevertheless, we can neglect this term when the gradient in volume fraction of these other species is small. Thus we only have to consider a force related to diffusion, and to the electric field, i.e., electromigration.

Frictional contributions are due to friction of the ion with other ions that have a different velocity (otherwise there is no friction), with the water, and with the membrane structure. Friction with other types of ions depends on the concentration of those other species, i.e., the more of them, the more friction, but the friction with the water, and with the membrane, these terms have no concentration-dependence. We neglect ion-ion friction now –otherwise see ref. [41]– so the friction of ions is only with the water that also flows through the porous medium, and with the membrane. Each frictional term is the product of a friction factor, related to the ion and that other phase, or species, and related to the velocity difference. We then arrive at (p. 128 in ref. [25])

$$\mathcal{F}_{\text{friction},i} = -\sum_j f_{i-j}^* (v_i - v_j) = -f_{i-w}^* (v_i - v_w) - f_{i-m}^* v_i \quad (88)$$

where the water velocity is v_w and we included in the last part that the velocity of the membrane is zero, $v_m = 0$. We present here velocities as superficial velocities, per unit of total membrane area. In a more detailed analysis we start with interstitial velocities, for instance for the water that is its velocity flowing through the space not occupied by the solutes in the pores of the membrane structure. In that more detailed analysis we also first set up these forces balances in the direction along the flow path of a tortuous pore, and when converting to an x -coordinate that is directed straight across the membrane, a tortuosity factor arises in the final equations. These detailed steps go beyond the scope of this tutorial review, but are explained in ref. [26].

We can rewrite the ion-water frictional coefficient to a diffusion coefficient of the ion in the membrane, $D_{m,i}$, by the Einstein equation, $f_{i-w}^* = RT/D_{m,i}$ (p. 128 in ref. [25]). This diffusion coefficient is lower than in free solution because it includes the porosity and tortuosity of the membrane pores. We now obtain an extended Nernst-Planck equation for ion flow inside a membrane (Eq. (7) in ref. [4])

$$J_i = K_{f,i} c_i v_w - K_{f,i} D_{m,i} \left(\frac{\partial c_i}{\partial x} + z_i c_i \frac{\partial \phi}{\partial x} \right) \quad (89)$$

where we also included that $J_i = c_i v_i$. Concentrations here are those in the membrane pores, but we leave out the index m . The friction factor, or hindrance function, $K_{f,i}$ is given by $K_{f,i} = 1/\{1 + f_{i-m}^*/f_{i-w}^*\}$, and has a value between 0 and 1, and describes that ions have a direct friction with the membrane structure, not only with the water that directly envelopes them and flows past

them. The ion-membrane frictional coefficient is f_{i-m}^* but its value we do not need to know when we phrase the problem in terms of $K_{f,i}$. In a further extended theory that includes how part of the pore volume is occupied by solutes and not available for the flow of free water, then instead of the function $K_{f,i}$, we have two different functions that we can call $K_{c,i}$ and $K_{d,i}$, completely in line with literature on NF transport theory where often separate ‘convective’ and ‘diffusive’ hindrance functions are implemented. But in the two-fluid model, when we can assume that solutes do not occupy very much space in the pores, there is one friction function, $K_{f,i}$. If $K_{f,i}$ is unity, ions have no friction with the membrane matrix, and then the dependence of water flow on pressure simplifies significantly, with a constant water permeability factor, k_{F-m}^* , which otherwise depends on the local concentration in the membrane, as we will explain in the next section. In any case, Eq. (89) is a general expression for ion transport in porous media, extended compared to the standard Nernst-Planck equation with advection and with ion-membrane friction, which leads to the factor $K_{f,i}$.

5.3 Flow of water through membranes in the two-fluid model (TFM)

Now we continue with the transport of water through the pores of a porous medium. In the derivation that follows, we assume the pores are filled with water completely and ions are ‘point charges’. We also neglect the effect of tortuosity. For a full account, see Ch. 8 in ref. [26]. We can now also set up a balance of forces acting on a volume element of water in the pore, with the driving force a gradient in total pressure, i.e., $-\partial P^{\text{tot}}/\partial x$ is the force acting on a volume element of water (it indeed has the correct unit N/m³). The forces acting on the water are a friction with the membrane structure, $f_{F-m}^*(v_m - v_w)$, where $v_m = 0$ because the membrane is immobile (ref. [25], p. 126), and friction with all types of solutes, and this friction is proportional to the concentration of that type of solute, $\sum_i f_{i-w}^* c_i (v_i - v_w)$. All these forces are added up, and their sum-total is set to zero, resulting in

$$-\frac{\partial P^{\text{tot}}}{\partial x} - f_{F-m}^* v_w + \sum_i f_{i-w}^* c_i (v_i - v_w) = 0. \quad (90)$$

We can solve Eq. (90) at each position in the membrane, together with the extended Nernst-Planck equation, Eq. (89). Now, a significant simplification results when we can assume that there is no ion-membrane friction, and thus we can set $K_{f,i} = 1$ in the NP equation, Eq. (90), for all ions. Then inserting the NP equation in Eq. (90) results in

$$-\frac{\partial P^h}{\partial x} + \frac{\partial \Pi}{\partial x} - f_{F-m}^* v_w - RT \sum_i \left(\frac{\partial c_i}{\partial x} + z_i c_i \frac{\partial \phi}{\partial x} \right) = 0 \quad (91)$$

where we implemented that $P^{\text{tot}} = P^h - \Pi$ and $D_{m,i} = RT/f_{i-w}^*$. Now, because in the ideal case that we discuss here the osmotic pressure equals the total ion concentration times RT , thus $\Pi = c RT$ with $c = \sum_i c_i$, Eq. (91) simplifies to

$$-\frac{\partial P^h}{\partial x} - f_{F-m}^* v_w + X RT \frac{\partial \phi}{\partial x} = 0 \quad (92)$$

and thus we have an exact cancellation of the osmotic pressure acting on the water, and the diffusional forces acting on the ions, which transfer to the water via ion-water friction. But this only happens when the ions have no friction with other phases except for the water. In Eq. (92) we also included local electroneutrality in the membrane, $\sum_i z_i c_i + X = 0$ (with X either positive or negative). Then we end up with Eq. (92) where we have a hydrostatic pressure gradient acting on the water, water-membrane friction, and an electrostatic body force term, which actually entered this force balance because of water-ion friction. We can rewrite Eq. (92) to

$$v_w = -\mathbb{k}_{F-m}^* \left(\frac{\partial P^h}{\partial x} - X \frac{\partial \phi}{\partial x} \right) \quad (93)$$

where we made the replacement $f_{F-m}^* = 1/k_{F-m}^*$. For an uncharged membrane, $X = 0$, and then Eq. (93) simplifies to the Darcy equation, Eq. (35). Thus water flows across an uncharged membrane not by diffusion but because of an internal pressure gradient, in line with Pappenheimer in 1953 [21], Mauro in 1957 [43], Ray in 1960 [44], and Manning in 1968 [18]. Thus, Eq. (93) is an extension of Darcy's law, valid for charged porous media [37]. For a constant membrane charge, it can be easily integrated between the two sides of the membrane. Let it be reminded that the last three equations all assumed $K_{f,i} = 1$.

The final element of transport theory through a porous structure filled with a continuum fluid such as water, with dispersed enclosed entities, such as ions, is mechanical equilibrium at the outsides of the structure, where it interfaces with the outside solution. This was already discussed in Section 3. Here we derive the mechanical equilibrium condition based on Eq. (91), because with all gradient terms much larger in the thin interfacial region than the terms proportional to velocities (or in other words, the structure of the interfacial region is not significantly modified by flows of ions and water across it), we arrive at the conclusion that P^{tot} is invariant across any thin interface, and thus $P^h - \Pi$ is invariant as well [35, 37, 45], and this leads to the earlier conclusion that across an interface the change in osmotic pressure Π (either up or down) equals the change in hydrostatic pressure P^h across that same interface. Again, let it be reiterated that osmotic pressure is not really a pressure in the sense that it has a direct 'hydraulic' or 'mechanical' impact. It is simply a measure of solute concentration, in the ideal case linearly related to solute concentration.

So this allows us to address the question why water always has a tendency to flow to that side of a membrane that has the highest salt concentration. The origin is the two osmotic pressure changes across the two outer surfaces, and they are different on the two sides of the membrane if the outside solutions are different [43, 46]. And both for the case with neutral solutes and an uncharged membrane, and for the case with ions and a charged membrane, in both cases this leads to a hydrostatic pressure profile in the membrane that is such that it pushes the water to the concentrated side [44]. For a neutral membrane the osmotic pressure goes down the most on the high-concentration side, and thus hydrostatic pressure there drops the most as well (from a value outside the membrane). For a highly charged membrane, osmotic pressure goes up upon entering the membrane, but because inside the charged membrane osmotic pressure is relatively constant, at a value just above the charge density $|X|$, this increase in pressure is the lowest on the high-concentration side, and thus again we develop a pressure profile inside the membrane that pushes the water to the high-salinity side. All of this assumes the two outside hydrostatic pressure to be the same, or close to one another. If not, if we apply a hydrostatic pressure difference across the membrane, we can overcome the osmotic effect, and this is of course what we do in RO and NF. For more details see Ch. 8 in ref. [26]. We quantify these forces in more detail for a charged membrane in the next section.

5.4 Counterfluxes of salt and water in osmosis

To illustrate the two-fluid model, and to provide evidence for its validity and accuracy, we provide results of an accurate fit of this theory to novel data of the flow of water and salt across an ion-exchange membrane. This experiment is called osmosis, and water and salt flow because of a difference in salt concentration between the two outside solutions on either side of the membrane, without pressure differences or current. Results of this experiment are presented in Fig. 11 as function of the concentration on the low-salinity side, c_d , which has values changing (in time) from 4 mM to 130 mM. During that time the salt concentration on the concentrate side decreases from

800 to 525 mM. Over the 35 hr of the experiment, the volume on the diluate side drops by 40% and that water flows to the concentrated side. For details see Ch. 8 in ref. [26]. In this experiment water flows to the high-salinity side, while salt flows in the other direction. Thus, inside the membrane these two flows are in opposite direction. This makes this an experiment that is not easy to ‘get right’ with a theoretical model, i.e., it is a very good test of a mass transfer theory. Note that to the best of our knowledge, such detailed data was not published before, let alone described accurately by theory that simultaneously describes water and salt flows.

Now, it turns out that the mass transport theory described above, with cation and anion fluxes the same, because the electrical current through the membrane is zero, and with equal diffusion coefficients of the ions in the membrane, works excellently and results in the very good fit to the data shown in Fig. 11. In the integration of the water force balance, Eq. (90), the total concentration of ions in the membrane is set to a constant value, which is calculated as an average of the values on the two sides, and these concentrations are calculated by Eq. (76). The salt flux is given by Eq. (97) extended for diffusion which is a term $-\frac{1}{2}k_m\Delta c_{T,m}$, where Δ refers to a difference in $c_{T,m}$ between the two sides of the membrane. Based on ref. [47] we derive a value of $\Phi_i=0.82$. The partitioning function Φ_i is part of the calculation of $c_{T,m}$, by making use of Eq. (85) to be evaluated for both ions, leading to the equation just above Eq. (98). Membrane charge density is $X=-5.1$ M and membrane thickness is $L_m=120\text{ }\mu\text{m}$. To fit the model to the data, we use for the hindrance function $K_{f,i}=0.51$. Because $K_{f,i}\neq 1$, we cannot use the Schlögl-equation, Eq. (93). The values we derive for the transport parameters $k_m=0.31\text{ }\mu\text{m/s}$ (corresponding to a diffusion coefficient in the membrane about $50\times$ less than in free solution), and $k_{F-m}^*=55\text{ fm/Pa.s}$ (which corresponds to $\sim 20\text{ mL/m}^2\text{/hr/bar}$) are in line with earlier reported values in ref. [41].

Thus, we can conclude that the two-fluid model including chemical and mechanical equilibrium is very well capable of describing fluxes of water and salt in the osmosis experiment. Note again that these fluxes were in opposite directions inside the membrane. It is interesting to make a theoretical estimate of the velocities of these species, for which we assume a porosity of 34% and we consider velocities at straight angles across the membrane. We consider the moment in time when we still have the original concentrations on the two sides of 4 mM and 800 mM. The water velocity is now $\sim 0.3\text{ }\mu\text{m/s}$. The counterion velocity is constant across the membrane and is $\sim 13\text{ nm/s}$ (in the direction opposite to the water flow). The coions have a concentration that drops by a factor of $\sim 4\cdot 10^4$ between the concentrated and dilute side, and to have constant flux, their velocity increases by that same factor, from $\sim 0.8\text{ }\mu\text{m/s}$ to $\sim 31\text{ mm/s}$.

Thus, in the most extreme case, coions have a velocity of the order of $2.4\cdot 10^6$ times larger than the counterions. One can imagine that therefore an ion-ion friction may be a force that we need to include. And we may want to analyze if the rapid acceleration of coions across the membrane must be included in the force balance on an ion, Eq. (84). One also can wonder if chemical equilibrium is still valid for the coions that exit the diluate side of the membrane at this incredible velocity.

6 Extended models for reverse osmosis and electrodialysis

6.1 2D RO model for electrolyte solutions

For RO with salt solutions, much work still needs to be done both on the 1D and 2D level. In this section we present a simple 2D module calculation for an ideal 1:1 salt. In the future this must be extended to solutions of more ions, and effects of pH on ion protonation and membrane charge (acid-base reactions) must be included [8, 42, 48–51]. In the present section we only discuss a module calculation for a simple 1:1 salt of two ions of equal diffusion coefficient, equal Φ_i , for

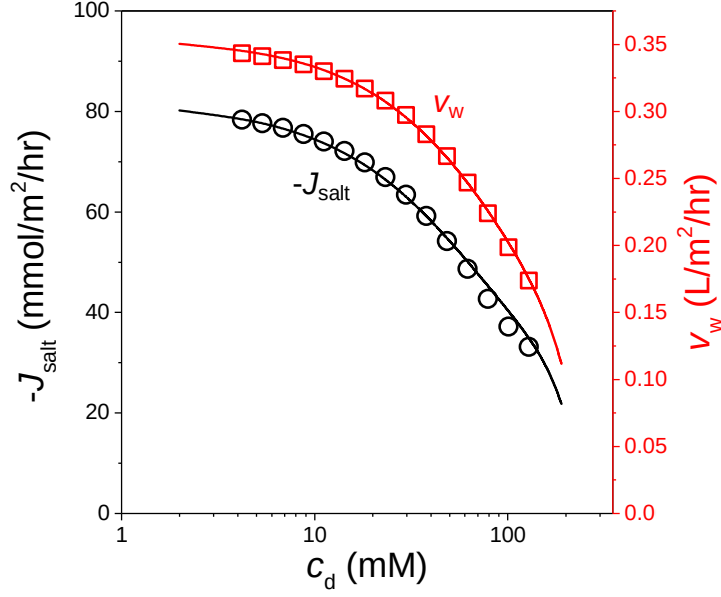


Figure 11: Experimental and theoretical results of salt flux, J_{salt} , and water velocity, v_w , across a cation-exchange membrane placed between two solutions with different salt concentration (initially $c_d = 4$ mM and $c_c = 800$ mM). Water flows to the high-salinity side, and salt in the other direction. There is no external pressure difference and there is no current. For membrane charge, thickness, porosity, diffusion coefficients, see main text.

a fixed membrane charge, in a transport model that neglects diffusion, similar to the RO model discussed in Section 3 for neutral solutes. Thus for RO with salt solutions, there is advection and electromigration. This line of modeling was already proposed for salt solutions by Dresner in 1972, ref. [19]. When compared with detailed RO modeling that also includes diffusion, the advantage of leaving out diffusion is that we do not need to solve ordinary or partial differential equations for transport of ions across the membrane, though of course the simplified model is potentially less accurate. Models that do include diffusion for RO of salt solutions are given for a symmetric 1:1 salt, in a 1D calculation, by Eqs. (37)-(39) in ref. [40] based on the work of Schlögl (1964). Starov and Churaev [4] include the interplay of membrane charge and non-electrostatic partitioning, but do not consider the relationship between water flowrate and pressure. When we assume membrane charge to be much higher than the salt concentration outside the membrane (not necessarily a valid approximation in RO), then Eq. (41) in ref. [40] results. Included is Donnan equilibrium at the two outsides of the membrane, which we will later extend to include a non-electrostatic contribution to ion partitioning, by including a Φ_i -factor.

In the limit of high transmembrane water flow velocities, v_w , salt flux is proportional to v_w , see Eq. (44) in ref. [40]. The flux of water across the membrane is described by an equation similar to Eq. (93), again driven by a hydrostatic pressure difference minus an osmotic pressure difference, with a frictional prefactor that relates both to the water-membrane permeability, and to the membrane charge density (which is approx. the counterion concentration). This latter effect arises from the friction of water with the almost stagnant counterions.

Just presented was a short description of the results of 1D modeling of RO with a 1:1 salt when diffusion is still included. To make these equations somewhat more tractable, it was necessary to assume in ref. [40] that X was large. We now continue with a model based on the limiting case that diffusion can be neglected. Then we do not have to make the assumption that X is very large (relative to salt concentration) to arrive at a quite elegant model. Thus for each ion we now have a

flux described by

$$\frac{J_i}{K_{f,i}c_{m,i}} = v_w - z_i D_{m,i} \frac{\partial \phi}{\partial x}. \quad (94)$$

Because diffusion is neglected, the result of solving Eq. (94) is that concentrations are invariant across the membrane. Then also $\partial \phi / \partial x$ is a constant, i.e., the potential changes linearly. If a calculation is made including some diffusion, one notices that these assumptions are correct except for a small region on the downstream side, where concentrations strongly drop to much lower values over a very short distance.

Without diffusion, for a solution with two monovalent ions, the zero current-condition, i.e., $I/F = \sum_i z_i J_i = 0$, can be combined with Eq. (94), which leads to

$$\frac{\partial \phi}{\partial x} = - \frac{X K_{f,i} v_w}{D_{m,+} c_{m,+} + D_{m,-} c_{m,-}} \quad (95)$$

where we set $K_{f,i}$ to the same value for both ions, and we implemented that in the membrane we have local electroneutrality, $c_{m,+} - c_{m,-} + X = 0$, where X is the membrane charge, which can be negative and positive. We insert Eq. (95) in Eq. (94) and integrate across the membrane for both ions, implementing that $c_{m,i}$ is constant (equal to the value at the upstream side). The flux of cations is equal to the flux of anions, thus we can add these two expressions, divide by 2, and then obtain for the salt flux

$$J = \frac{1}{2} K_{f,i} v_w \left\{ c_{T,m} + \frac{D_{m,+} c_{m,+} - D_{m,-} c_{m,-}}{D_{m,+} c_{m,+} + D_{m,-} c_{m,-}} X \right\}. \quad (96)$$

When we assume the two diffusion coefficients to be the same, then Eq. (96) simplifies significantly, to

$$J = \frac{1}{2} K_{f,i} v_w \left\{ c_{T,m} - \frac{X^2}{c_{T,m}} \right\} \quad (97)$$

where the total ions concentration in the membrane (cations plus anions) is $c_{T,m}$. Now, in Eq. (97) we can implement the Donnan equilibrium including the partitioning function (which relates to all non-Donnan effects), Eq. (85), which for a 1:1 salt results in $c_{T,m}^2 = X^2 + (2c\Phi_i)^2$ (we assume the same Φ_i for both ions), and we then obtain

$$J = \frac{1}{2} K_{f,i} v_w (2c\Phi_i)^2 / \sqrt{X^2 + (2c\Phi_i)^2} \quad (98)$$

where c without any index is the salt concentration on the upstream side of the membrane, which in a 2D model is z -position dependent, i.e., changing from inlet=feed to exit=retentate. If the membrane charge density X is much larger than $c\Phi_i$, then Eq. (98) simplifies to

$$J = K_{f,i} v_w 2c^2 \Phi_i^2 / X \quad (99)$$

which extends Eq. (44) in ref. [40] by including $K_{f,i}$ and Φ_i^2 . Thus we now derived a more general result, Eq. (98), for the same situation as in ref. [40] of a 1:1 salt with the two ions having the same diffusion coefficient, same Φ_i and same $K_{f,i}$, but now valid irrespective of the ratio of c over X , which is important in RO because X is not very large in RO, and c can be. For a module with high water recovery, thus strongly increasing salt concentration on the upstream side, this is a very useful extension. In the limit where $X \ll 2c\Phi_i$, then we obtain $J = K_{c,i} \Phi_i c v_w$, i.e., the same result as Eq. (30) for neutral solutes.

The transmembrane water flowrate, v_w , is given by Eq. (93), in which we implement Eq. (95). By assuming equal diffusion coefficients, we can rewrite the resulting equation to

$$v_w = +k_{F-m}^* \Delta P^{h,m} / \left(1 + \frac{k_{F-m}^* R T X^2}{D_m c_{T,m}} \right) \quad (100)$$

where in the present model without diffusion, the concentration in the membrane is evaluated as that at the upstream side of the membrane. As before, we define Δ as a value on the upstream side minus on the permeate side, both just in the membrane. These hydrostatic pressures just in the membrane, as required in Eq. (100), are equal to the hydrostatic pressures just outside the membrane, $P^{h,\infty}$, on that respective side, plus the change in osmotic pressure across the interface. On the upstream side, this difference is $\Delta\Pi = RT(c_{T,m} - 2c)$ (defined just inside the membrane minus just outside), with the expression for $c_{T,m}$ given above Eq. (98). On the downstream side, we use the calculated c_p based on which $c_{T,m}$ on that side of the membrane is calculated by the same equation. Now, with a charged membrane, and if Φ_i is not too low, then the two $\Delta\Pi$'s (one on either side) are positive numbers (while they were negative for RO with neutral solutes). We merge the entire denominator of Eq. (100) in a modified permeability, and we also include the pressure drop in a support layer located just below the top layer. This leads to the permeability being replaced by a general k^* , with a lower value, as if the top layer itself is thicker. We then simply end up with $v_w = k^* \Delta P^{h,m}$.

We implement these equations in a 2D model for an RO module similar to the approach for neutral solutes, where we solved transport equations at each position ξ in the module, which led to the calculation output described by red circles in Fig. 6. We use the same approach and analyze a module with given membrane area A , and always the same permeate production (productivity, SP) that we set to a realistic value of 10 LMH (liters permeate water per square meter per hour) with a feedwater salt concentration c_f either of 50 mM or 500 mM, representative of moderately brackish water, and of seawater, respectively. At $c_f = 50$ mM, the minimum pressure is $\Delta P_{\min}^{h,\infty} = 16.2$ bar, which is the required pressure to just push the flowrate of 10 LMH through the membrane at a water recovery approaching zero, i.e., on the entire upstream side, the water has a concentration equal to that of the feed. This situation may give us the lowest permeate concentration, see Fig. 12A, but also a very high cost of pressurizing the very large feedwater stream, see Fig. 12B. When we increase the applied pressure we can increase water recovery (have to flow less feedwater into the module, still producing 10 LMH permeate), while c_p only changes moderately. The total energy investment decreases and thus the total cost of operation, TC, goes down, until a minimum is reached beyond which TC goes up again. We use Eq. (51) to describe the cost of operation of the RO module. Because of the fixed system size, the second part related to membrane costs is constant, and thus, to find the optimum pressure and other factors, various cost factors such as CUE are irrelevant as well. The only thing of importance in the present calculation is the applied pressure, $\Delta P^{h,\infty}$, times the feed flow rate $\phi_{v,f}$ (which is proportional to the fixed production flowrate, divided by WR) and times the factor $1 - \eta_{ERD}(1 - WR)$, in which we use $\eta_{ERD} = 0.5$. We leave out other cost factors relating to pretreatment or brine discharge.

Interestingly, for $c_f = 50$ mM, this simple calculation provides a minimum in total costs at a pressure around 21 bar (about 30% above the minimum pressure), and a water recovery at that condition of $WR \sim 78\%$. For seawater, $c_f = 500$ mM, the minimum pressure is 38.2 bar, the optimum is at 53 bar (about 35% beyond the minimum) and the water recovery at that most optimal condition is 50%. This calculated value of water recovery for seawater desalination is actually a value common for operation of RO seawater desalination plants. The values of X and Φ_i that we use lead to a permeate concentration of $c_p = 0.20$ mM and thus a retention of $\sim 99.9\%$ for $c_f = 50$ mM (at the optimum) while for seawater ($c_f = 500$ mM) we have at the optimum a permeate concentration of $c_p \sim 8$ mM, and thus a retention of $R_i = 98.4\%$, which is a realistic number for seawater RO. Interestingly, for brackish water, retention is much better, more than $\sim 15\times$ better than for simulated seawater (when we define 'goodness' as how close we are to 100%).

There are analytical solutions for the minimum pressure, which is the pressure we need in the

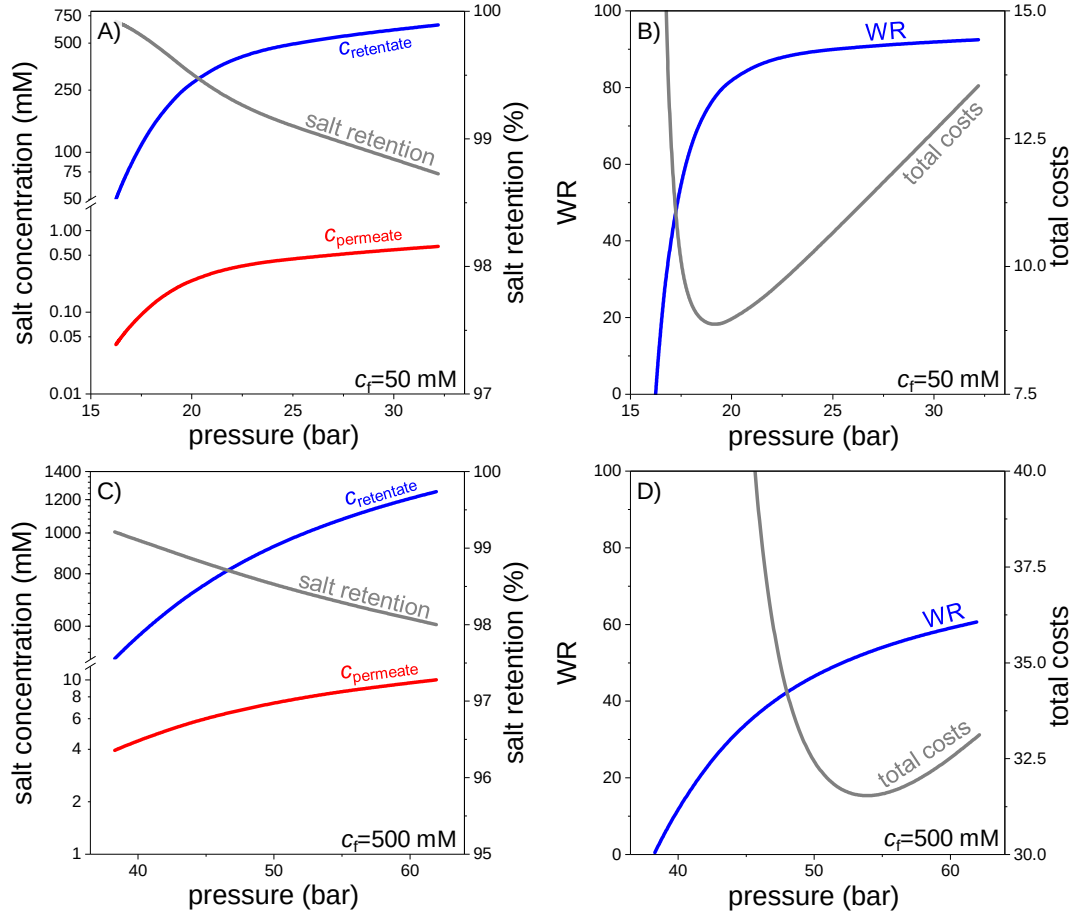


Figure 12: Calculations for a 2D RO module based for fixed water productivity of 10 LMH for a 50 mM 1:1 salt solution as feed (A and B) and a 500 mM solution (C and D) ($k^* = 2.0$ pm/s/Pa, $\sigma_i = 0.96$, $X = 0.2$ M).

limit of very low WR (similar to the 1D modeling of Section 3) and for the permeate concentration in that limit. The expression for minimum pressure is not very compact, but the limiting (minimum) permeate concentration $c_{p,\min}$ is simply given by J_i from Eq. (98) divided by v_w , with the upstream salt concentration c replaced by the feed salt concentration, c_f . For parameter settings given in Fig. 12, we obtain $c_{p,\min} = 3.92$ mM for 500 mM feed concentration, and for 50 mM feed salinity, this minimum permeate concentration is about $100\times$ lower.

6.2 2D cross-current ED model

We continue with advanced calculations for electrodialysis (ED). In Section 4 a model for ED was discussed where the solutions flow co-currently along the membranes. [Note, the ‘current’ in this description of flow patterns does not relate to electrical currents, but to the direction of water flows in the channels.] In the present section we compare this flow pattern with two other modes of water flow. The first alternative is operation in counter-current flow as explained in Fig. 13, mathematically similar to co-current flow, but now the direction of water flow in the concentrate channels is in opposite direction to that in the diluate channels. Finally we compare with cross-current flow, where flow in the concentrate channel is ‘at a 90 degrees angle’ relative to the flow direction in the diluate channel. Physically, a three-dimensional (3D) problem is solved. We compare ED operation in these three modes on the basis of the important curve of efficiency η_{ED} vs. specific productivity,

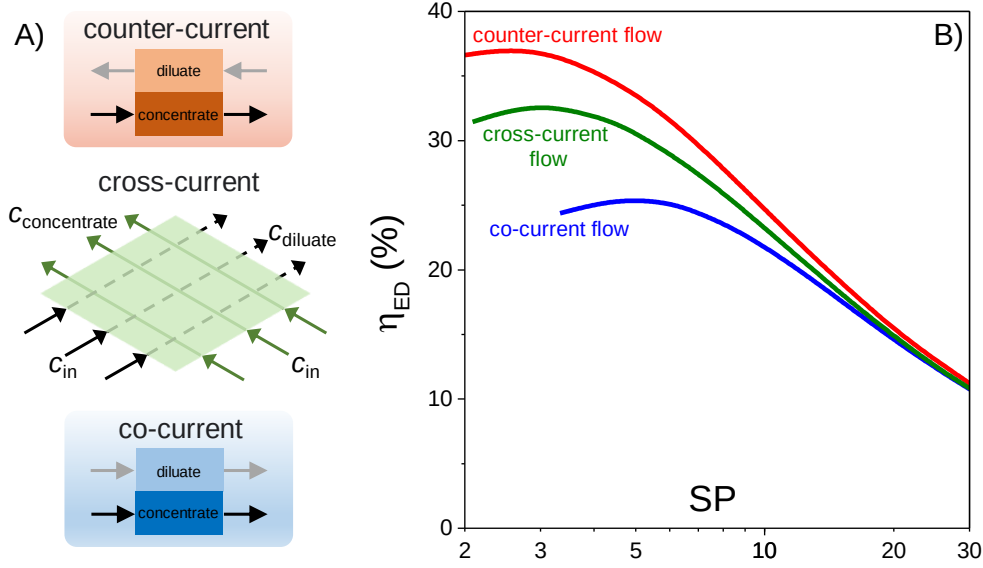


Figure 13: Comparison of three flow patterns in an ED module, namely counter-current, cross-current, and co-current flow. Calculation results for an ED module based on input parameters from Section 4, evaluated as efficiency η_{ED} versus specific productivity, SP. All calculations for desalination from 500 to 50 mM, at a water recovery of 50%.

SP. So each calculation is made for the same separation, where the diluate concentration goes down from 500 mM to 50 mM, and water recovery is 50%. As Fig. 13 demonstrates, efficiency is best (for a certain value of SP) in counter-current operation, next best is cross-current, and last is co-current flow. The differences are significant in a range of SP-values between ~ 3 and 10. And in addition, because it is never useful to work 'left of' the maximum, for co-current flow one would not operate below $SP \sim 5$, while for counter-current flow it can still be sensible to operate as low as $SP = 3$. Thus, in these regards, there is a significant effect of the flow geometry on operational efficiency. Nevertheless, for high production rates, in this case beyond $SP \sim 15$, the three flow geometries all give the same result, i.e., the exact type of flow does not matter at these high flow rates. The flow geometry only matters at lower flow rates (low SP), and as discussed before, these condition are optimal when material costs are favourable relative to energy costs [31]. In the other limit, when costs of materials are a very large part of the total operational expenses, we will operate with less membrane area, at higher electrical currents, and then calculation results are that the dependency of efficiency on productivity does not depend very much on the mode of operation.

6.3 Optimization of two equal ED modules operating in cross-current stacked differently

Now we extend the calculation for ED to the multi-module level. A layout is considered with two ED modules (two ED stacks) that can be different from one another, as a whole forming the complete ED system. We consider different ways to *stack* these two modules in the system. We make comparisons for the same value of SP evaluated at the system-level. Thus, we add together all areas in the two modules to a total area A that is used in the calculation of SP, and if we compare with a system consisting of a single module, that one module can have more area to end up at the same system-level SP in this comparison. Results of this calculation are presented in Fig. 14. All calculations in this section are based on the most realistic module layout (flow pattern) which is cross-current flow. Thus calculations here are based on a 3D model for each separate module, and

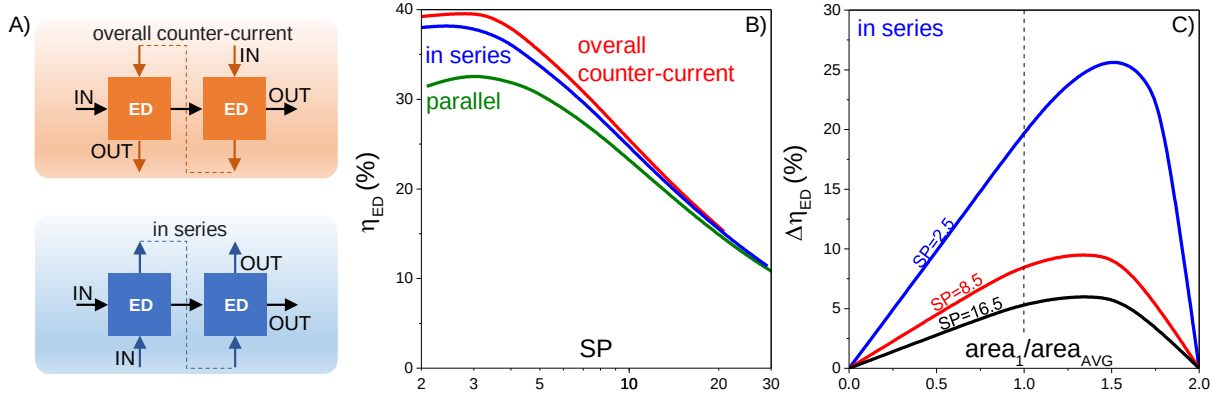


Figure 14: Comparison of different methods of stacking two ED modules. Each individual stack is described by the 3D cross-current geometry. The parallel mode refers to the two units operated in parallel, or equivalently, as one larger module. The other two modes are explained in A). In B) the two stacks have the same size, while in C) for the in-series flow mode at three values of overall SP, we give the percentage-wise increase in η if we work with different sizes of the two stacks.

then two modules are jointly considered, and for each value of SP, all free parameters are varied to arrive at an optimized efficiency η_{ED} , such that the curves in Fig. 14 are optimal for any chosen mode of stacking.

In Fig. 14 the line for single module is the same as if we had two smaller modules each with half the area, placed next to one another (in parallel). Thus the ‘parallel’ line in Fig. 14 is the same as the line for cross-current flow in Fig. 13. The second line is for the two modules placed in series, one behind the other. The cell pair voltages in both modules can be changed, and the aim is to arrive at the required setpoint (desalination from 500 mM to 50 mM) with the lowest total energy investment. Finally, the upper line is for two modules that on the system-level are operated in counter-current (again, for all calculations here each module by itself is operated in cross-current). Clearly, at moderate SP-values we can increase efficiency by either of these system-level stacking options. Around an SP of ~ 3 an improvement by about $\sim 25\%$ is possible, when we go from one module to two (smaller) modules. In each module the cell voltage must be chosen exactly right to obtain this advantage, otherwise efficiency is actually worse than just having a single module.

In a next set of calculations, presented in Fig. 14C, we only analyze the ‘in-series’ system-geometry and change the sizes (i.e., membrane area) of the two modules, but still with each curve at the same system-level SP (i.e., if one module is made larger, the other must be made smaller), while we also optimize the two voltages in each module. Reported is the *relative* increase in efficiency relative to the situation of a single module. The two extremes, at the very left and very right in Fig. 14C refer to the single module, and the condition in the exact middle is the in-series geometry with two equally-sized modules. We notice that we can gain a further increase in efficiency if the first unit is increased in size by up to 50% and the second one (thus) made smaller by that same amount. Note that this change in size can simply be realized by having more cell pairs in the first, larger, unit, and less in the second, smaller, unit (with all water still flowing through both stacks), which increases the residence time in the first unit and reduces it in the second. At the lowest SP (upper curve), we can have a relative efficiency increase (relative to the single module) of $\sim 25\%$. The improvement compared to the case of two equally-sized modules (at vertical dashed line) is relatively modest, not more than about 5%, but it might pay off the effort to increase the number of cell pairs in the first unit, and reduce this number in the second unit.

6.4 3D cross-current ED model with concentration polarization

In this final section we further improve the calculation for an ED module by including concentration polarization (CP). We make calculations without CP, like the model of the previous section, thus where at each macroscopic position in the module we have across the channel one salt concentration, i.e., across the width of the channel, perpendicular to the water flow direction in the channel, we assume perfect mixing ('crosschannel mixing'). These calculations are compared with results of a full 3D cross-current model that includes CP. Parameters that we use for this calculation are a charge density of $X=4.0$ M, a membrane transfer coefficient for the ions of $k_m=0.24$ $\mu\text{m/s}$, a channel thickness of $L_{\text{ch}}=200$ μm , and an ion diffusion coefficient in the flow channel of $D_{\text{ch}} = 1 \cdot 10^{-9}$ m^2/s , which includes a porosity and tortuosity effects. We do not include dispersion, and thus solve the Nernst-Planck equation across the channel using this one value for D_{ch} . In the numerical calculation, based on extensive discretization in all directions into a large number of gridpoints, water and salts at a certain distance from the membrane is advectively passed on along the membrane in the required direction. And there is no averaging or the like of the salt concentration profiles going from one gridpoint to the next. We assume plug flow for the water flow rates, i.e., at each position the water flow rate in the direction along the membranes, is the same. We neglect water flow through the membrane, and in that regard the membranes are ideal. However, we do include that the membranes are leaky for coions, as discussed at length in Section 4, and this turns out to be a very important aspect. We assume steady-state in an ED cell with unsegmented electrodes, i.e., at each position in the stack there is only one cell pair voltage, which is the same as the approach in previous sections.

In many cases we notice that significant concentration profiles develop across the width of the channel, sometimes up to a factor ~ 4 difference in concentration between the center of a channel and the channel/membrane interface. Because of these concentration profiles the resistance across the diluate channel goes up somewhat but this is a small effect. Much more important are two other effects related to the membranes. First of all, CP increases the Donnan potential on the diluate/membrane interface (more than it reduces the Donnan potential on the concentrate/channel side), because concentrations at the channel/membrane interface are now much lower. For this reason the cell pair voltage goes up and thus the energy consumption. But another effect is even more important. Because of the decreasing concentration in the diluate channel at the channel/membrane interface, and likewise the increasing concentration on the concentrate side, salt leakage from the concentrate channels significantly increases, which lowers the current efficiency, λ . Now, because we need to arrive at the same objective function of a certain salt removal, to compensate for this leakage, currents must be increased, and this also leads to higher voltages. This increased leakage, and the required compensation for it such that we arrive at the same objective function, is the key reason why if we include CP in a model in which membranes are leaky for co-ions, we obtain a significantly larger energy consumption, in some of our calculations up to a factor of 2.5 more.

Calculations are made for many values of the specific productivity (SP), and for a required salt removal, i.e., a certain reduction in salt concentration achieved in the diluate channel. Thus in each calculation the cell pair voltage is adjusted until the required salt removal is achieved. Based on each calculation the average current is determined and the energy efficiency. We then calculate the energy ratio, which is defined as the energy efficiency from a calculation using an ED model with CP divided by the energy efficiency in an ED model without the CP effect. This number describes the energy input required according to a model with CP (the real, or at least more accurate, energy input), relative to the energy input calculated in a model that neglects CP

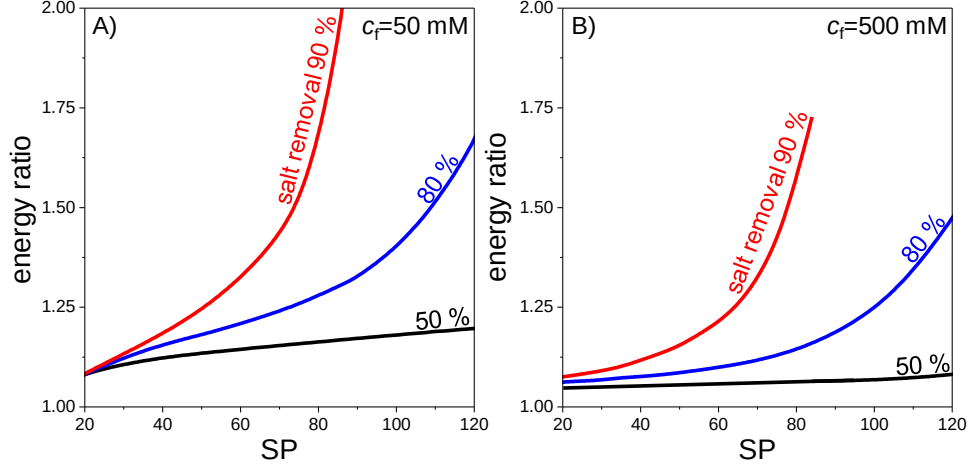


Figure 15: Calculations for ED with 3D cross-current geometry and non-ideal membranes, with and without concentration polarization (CP). For two values of feed salt concentration and three values of salt removal (percentage-wise reduction in diluate salt concentration), results are presented for the energy ratio as function of specific productivity, SP. The energy ratio describes the energy use in the ED stack (to reach the required salt removal) when we do include CP (which is a more precise model) over a model that neglects CP (which is a model based on the assumption of ‘crosschannel mixing’).

(based on perfect ‘cross-channel mixing’). One can also conceive of it as the technological potential to reduce energy consumption by developing ED channel designs that reduce the CP effect. Thus we present in Fig. 15 results of the energy ratio, for different values of salt removal and feed salt concentration, as a continuous function of SP. These energy ratios are always larger than unity, and this means that an ED module including CP always needs more energy than without CP to arrive at the same salt removal. As Fig. 15 shows, for low productivity, and moderate salt removal (50%), the energy ratio is close to unity, especially for 500 mM feed salt concentration (see panel B at low values of SP). At a low salt removal of 50%, the energy ratio stays rather constant with increasing SP, for both feed salt concentrations. However, this is very different for a higher salt removal, where calculations show energy ratios going up to values beyond an energy ratio of 2, i.e., the effect of CP on the energy consumption is low at moderate salt removal, but at high SP and high salt removal (above 80%), CP starts to play an important role and the energy ratio goes up significantly. Thus, both for low and high feed salt concentration c_f , the effect of CP can be very significant when we simultaneously strive for a high salt removal and high specific productivity.

7 Conclusions and Outlook

In conclusion, we have presented theory for reverse osmosis and electrodialysis, demonstrating their background in fundamental transport theory, and illustrating their application in simple 1D geometries, modules and even for system optimization of multiple combined units. For RO we presented a more realistic model than the solution-diffusion model and identified that the advection-only limit is likely a very relevant simplification. In a 2D module, an analytical result was derived that leads to a simple model of RO for neutral solutes, and leads to a further simplification (the $\langle c \rangle$ -model) valid at high productivity. For ED we develop new equations for Donnan equilibrium that extend the standard ideal expression. We present analytical equations for current efficiency, showing that this is a process parameter, not a membrane material property. We analyze combina-

tion of multiple modules in a system, and show that for RO there is no positive effect if upstream pressure losses (from entrance to exit) are neglected. For ED, it can be useful to couple multiple modules, but only if membranes are relatively inexpensive and we can work at high efficiency. Concentration polarization in ED has a negligible effect on the energy consumption for high feed water concentrations (seawater) and salt removal below 80 %, while CP should be considered for deep desalination of water with low feed salt concentrations.

As an outlook, let us summarize a few topics that we did not address. First of all, it is important to study how to extend both in RO and ED from simple 1:1 salt solutions to multi-ionic solutions. Even the addition of one extra type of anion or cation significantly changes the entire modeling framework. In addition, in real water sources also the protonation degree of ions must be considered, which depends on local pH. At high concentrations, ions also associate in ion pairs. These effects are of utmost importance because for instance an ammonium ion is acted on by the electrical field, but the ammonia species, is not. Thus retention of these ions is strongly pH-dependent. For very tiny pores, related is the effective size of ions, that has an impact on their partitioning and their mobility within the membranes. Ions that have a higher charge, have captured more hydration water, are expected to be slower. Important theory for acid-base reactions between ions in solution assumes that these reactions are infinitely fast, but it is interesting to study if that is a correct assumption. Another important assumption is local electroneutrality in channels and in membranes. But especially in reverse osmosis where membranes are only for instance 100 nm thick, it would be interesting to know if Poisson's equation must be used to replace local electroneutrality across the complete membrane.

Other topics of significant relevance to study are the use of NF and RO to remove organic micropollutants (OMPs) and other charged molecules such as perfluoroalkyl substances (PFAS). A very different topic is the theoretical study of electro-deionization (EDI), which is an ED system with channels filled with (mixed bed) resin particles. It is quite challenging to understand how this modification helps to reduce energy costs and to obtain ultra-pure water. The explanation might relate to water-splitting reactions where acidic and basic resin particles touch.

Nomenclature

Symbols		
a	Specific membrane surface area, $a = 1/L_{ch}$	m^{-1}
A	Membrane area	m^2
$\mathcal{A}$	Specific membrane area, i.e., the total membrane area per m^3/s of freshwater to be obtained	s/m
$\langle c \rangle$	Average upstream concentration of solutes	mol/m^3
$c_{i,j}$	Concentration of solute or ion i in stream j	mol/m^3
$c_{m,i}$	Concentration of ion i in membrane	mol/m^3
$c_{T,m}$	Total concentration of two ions together at any position in the membrane	mol/m^3
CMP	Material costs	a.m.u./u.o.e.
CUE	Costs of a unit of energy	a.m.u./u.o.e.
$D_{m,i}$	Membrane-based diffusion coefficient, $D_{m,i} = \varepsilon D_{\infty,i}$	m^2/s
e_{min}	Theoretical minimum energy of desalination	W
e_{min}^c	$e_{min}^c = E_{min}/\Pi_f$	
E_{min}	Theoretical minimum energy of desalination per m^3 of freshwater produced	J/m^3
f	Dimensionless z -dependent upstream flowrate, $f = \phi_v(z)/\phi_{v,f}$	
F	Faraday constant (96485 C/mol)	C/mol
$\mathcal{F}_{driving,i}$	Driving force on ion i	J/mol/m
$\mathcal{F}_{friction,i}$	Friction force on ion i	J/mol/m

$f_{\text{elec},j}$	Contribution of electrostatic Coulomb interactions between ions to the free energy density of a solution	J/m^3
$f_{\text{F-m}}^*$	Friction water with the membrane structure	Js/m^5
f_{i-j}^*	Friction factor of ion i with another ion j	Js/mol/m^2
f_{i-m}^*	Friction factor of an ion with the membrane	Js/mol/m^2
f_{i-w}^*	Friction factor of an ion with water	Js/mol/m^2
I	Current or current density	A , or A/m^2
J_i	Transmembrane molar flow rate of solutes (**)	$\text{mol/m}^2/\text{s}$
k_B	Boltzmann's constant ($1.3806 \cdot 10^{-23} \text{ J/K}$)	J/K
k_{ch}	Channel mass transfer coefficient, $k_{\text{ch}} = \varepsilon D_{\infty}/L_{\text{ch}}$	(m/s)
$k_{\text{F-m}}^*$	Water-membrane permeability, $k_{\text{F-m}}^* = \mathbb{k}_{\text{F-m}}^*/L_{\text{m}}$	m/Pa.s
$k_{\text{m},i}$	Membrane mass transfer coefficient, $k_{\text{m},i} = D_{\text{m},i}/L_{\text{m}}$	m/s
$K_{\text{f},i}$	Friction function for solute-membrane friction	
$K_{\text{c},i}, K_{\text{d},i}$	Hindrance functions for convection and diffusion	
L_{ch}	Flow channel thickness	m
L_{m}	Membrane thickness	m
m_1	Multiplier for total energy use	
m_2	Multiplier related to other plant expenses	
N_{av}	Avogadro's number ($6.0221 \cdot 10^{23} \text{ mol}^{-1}$)	mol^{-1}
n_i	Ion concentration in numbers per volume	m^{-3}
p	Membrane porosity	
p^{h}	Hydrostatic pressure	Pa (bar)
P_i	Passage of ion or solute i , $P_i = 1 - R_i$	
Pe_i	Membrane Péclet-number, $\text{Pe}_i = v_{\text{F}}/k_{\text{m},i}$	
R	Gas constant (8.3144 J/mol/K)	J/mol/K
R_i	Retention (or rejection) of ion or solute i	
S_i	Solubility (same as Φ_i)	
S_j	Entropy associated with a certain water flow	J/s/K
T	Temperature	K
t^*	Time-on-stream, $t^* = z/v_z$	s
tc	Dimensionless total costs	
TC	Total costs per time of operation (a.t.u, arbitrary time unit)	a.m.u./a.t.u.
V_{cp}	Cell pair voltage	V
v_{F}	Transmembrane water velocity, water flow rate (**)	m/s
V_{T}	Thermal voltage, $V_{\text{T}} = RT/F$	V
v_z	Velocity of water through a channel, i.e., along membrane	m/s
WR	Water recovery	
x	Coordinate towards and across membrane	
$ X $	Magnitude of the membrane charge density	mol/m^3
z	Direction along membrane	
α	Dimensionless cost factor	
α	Prefactor in modified Bjerrum theory ($\alpha \sim 225 \text{ J m/mol}^{4/3}$ at T_{room})	$\text{J m/mol}^{4/3}$
α_1, α_2	Over-pressurization factors	
Δc	Extent of desalination (difference in salt concentration between feedwater and product water, $\Delta c = c_{\text{f}} - c_{\text{p}}$)	mM
$\Delta P^{\text{h},\infty}$	Hydrostatic pressure difference across the membrane	Pa (bar)
Δt	Operational time a membrane can sustain	
$\Delta \phi_{\text{ch},j}$	Dimensionless potential drop across channel j	
$\Delta \phi_{\text{Donnan},j}$	Dimensionless Donnan potential at the interface of the membrane with electrolyte in the diluate or concentrate channel	
$\Delta \phi_{\text{Donnan,tot}}$	Dimensionless total Donnan potential across a membrane	
$\Delta \phi_{\text{m}}$	Dimensionless potential drop across the inner coordinates of the membrane	
ε	$\varepsilon = p/\tau$	

η	Energy efficiency, ratio of theoretical minimum energy to achieve a certain desalination over the actual energy investment (also called TEE, thermodynamic energy efficiency)	
λ_B	Bjerrum length	m
λ_{cp}	Current efficiency	
μ_i	Chemical potential of an ion	J/mol
$\mu_{aff,i}$	Chemical potential of affinity of ions with a certain phase	J/mol
$\mu_{exc,i}$	Chemical potential of volumetric interactions between ions	J/mol
$\mu_{ref,i}$	Reference value relevant when ions are part of chemical reactions	J/mol
v_i	Molar volume of an ion	m ³ /mol
$v_i P^{tot}$	Pressure insertion term	J/mol
Π	Osmotic pressure of the solution	Pa (bar)
$\Pi_{dbl/mem}$	Osmotic pressure at interface diffusion boundary layer-membrane	Pa (bar)
σ_i	Reflection coefficient, $\sigma_i = 1 - K_{f,i} \Phi_i$	
τ	Tortuosity factor of porous structure in membrane = τ^2	
ϕ	Volume fraction of all ions together	
ϕ_v	Volumetric flowrate	m ³ /s
Φ_i	Partitioning coefficient (same as S_i)	
ξ	Dimensionless coordinate axis along the membrane from inlet to exit (0 to 1)	
ω	Sign of the membrane charge	
Subscripts		
c	concentrate (what in RO is: retentate)	
f	feedwater	
p	product water, or freshwater (what in RO is: permeate)	
∞	conditions outside of membrane, or salt concentration in solution for a $z : z$ salt	

(**) These fluxes across the membrane are defined per unit geometrical ‘outer’ area of a membrane, i.e., they are defined as superficial velocities, not interstitial. This means they are not defined for instance per cross-sectional area of the pores only.

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